



Cerro Tololo Inter-American Observatory ▶
Community Science and Data Center ▶
Gemini Observatory ▶
Kitt Peak National Observatory ▶
Vera C. Rubin Observatory ▶

Modular Detector Controller Architecture

Architecture Handbook

Author: Braulio Cancino Vera

Version: 0.2

Generated: September 9, 2026

Distribution: Public



This handbook is derived and non-normative; resolved Architecture Decision Records control.

Document Control

Publication status

Document title	Modular Detector Controller Architecture
Document type	Architecture Handbook
Reference code	Pending
Version	0.2
Generation date	September 9, 2026
Distribution	Public

Approvals

Reviewers and formal approvals will be added when applicable.

Approval names and dates are entered only through the formal document-control process and are never inferred from repository history.

Change record

Version	Date	Description
0.1	Not recorded	Initial architecture handbook with generated reference appendices.
0.2	2026-09-09	Passive or active continuity with a common electrical interface, F6 coverage, and optional direct local response to LOOP LOW; optional interlock-supply supervisor with required relay disable and F6 reporting; safety CPLD and watchdog options; optional use of standard analog rails and local conversion; shared-rail conditioning including protected +12 V and digital loads; application-specific Ethernet and ADC rates; evidence-based diagnostics and conclusive maintenance verification; updated explanatory diagrams; clearer operating actions and reorganized explanations; synchronized concept guide, example function board, build instructions, and clearer detector interface terminology; explicit optional input eFuses for main and function boards; armed host-supervision rationale and pre-arm configuration verification; consolidated management-clocking explanation and aligned operating scenarios; clearer clock-loss and watchdog evidence explanation, including disarmed protection; separate watchdog and clock-loss detection with optional refresh alignment and local fallback.

About This Handbook

This handbook explains the modular detector-controller architecture as a coherent system story for technically competent readers who are not already familiar with the project.

Resolved Architecture Decision Records in the source repository remain authoritative. All figures and tables are explanatory and non-normative unless explicitly identified otherwise. This handbook is a derived, non-normative explanation and does not replace ADRs, interface-control documents, board-design specifications, or operating procedures.

The project originated at NOIRLab. Official project documents provide organizational motivation and historical context; resolved ADRs control the current architecture when earlier descriptions differ.

The system concept guide in `integration/system_concept_guide.md` is the maintained explanatory content source. Handbook chapters expand it and are kept aligned manually; shared reference tables are generated from their authoritative sources.

Contents

Document Control	i
About This Handbook	ii
I Context and Overview	1
1 Origins and Guiding Ideas	2
1.1 From a Local Need to a General Architecture	2
1.2 Guiding Ideas	2
1.2.1 Plan for replacement	2
1.2.2 Share common functions and specialize detector electronics	3
1.2.3 Scale by adding modules	3
1.2.4 Make safety and predictable behavior system properties	3
1.2.5 Separate architecture from implementation	4
1.3 Why This Approach Is Practical Now	4
1.4 Scope of This Handbook	4
2 The System at a Glance	5
2.1 Board Roles	5
2.2 The Four Main Connections	6
2.2.1 Power	6
2.2.2 Safety and coordination	6
2.2.3 Timing	7
2.2.4 Communication and science data	7
2.3 A Normal Operating Cycle	7
II Architecture	8
3 Modular System Structure	9
3.1 A Stable Platform with Specialized Function Boards	9
3.2 Main Board	10
3.3 Function Boards	10
3.4 Example: Inside a Function Board	11
3.5 Backplane Board	11
3.6 Bridge Boards and Additional Backplanes	13
3.7 Passive Terminators and Unused Slots	13
3.8 Responsibility Boundaries	14
4 Power, Supervision, and Noise Architecture	15
4.1 What Power Is Available?	15
4.1.1 Protected bulk power	15

4.1.2	Common analog rails	16
4.2	How a Board Uses the Available Power	16
4.2.1	Local conversion and protection	16
4.2.2	Conditioning shared rails	17
4.2.3	Returns, connectors, and budgets	18
4.3	What Happens When Power Fails?	18
4.3.1	Complete input cutoff	18
4.3.2	Supervision and diagnosis	19
4.3.3	Local interlock power	20
4.3.4	Startup and common-rail faults	20
4.4	Predictable Converter Switching	20
5	Hardware Safety and Interlock Architecture	22
5.1	Start at the Detector-Facing Output	22
5.2	Three Safety Layers	22
5.3	The Shared Open-Drain Fault Bus	24
5.4	The Continuity Loop	24
5.5	Independent Fault Paths	25
5.6	Protection Must Survive the Failure It Detects	27
5.6.1	Power independent of the processor rails	27
5.6.2	A watchdog independent of the supervised clocks	27
5.6.3	Implementing the functions with a safety CPLD	28
5.7	Safe First, Diagnose Second	28
5.8	Trips Remain Latched Until Explicit Recovery	28
5.9	Finding a Damaged Fault-Output Path	30
5.10	Safety Scope	31
6	Operating Lifecycle and State Coordination	32
6.1	Distributed Coordination	32
6.2	The Four State Families	32
6.3	Safe Startup and Qualification	34
6.4	Configuration in IDLE	34
6.5	Distributed Arming	35
6.6	Acquisition Within RUN	35
6.7	Hardware-First Disarm	37
6.8	Fault Entry	37
6.9	Explicit Recovery	37
6.10	Armed Host Supervision	37
6.11	Timing Categories Belong in ICDs	38
7	Timing Domains and Synchronization	39
7.1	The Distributed Sequencer Clock	40
7.2	Point-to-Point Slot Distribution	40
7.3	Acquisition SYNC and Cycle Agreement	40
7.4	Independent Management Clocking	41

7.5	Clock-Loss Detection and Watchdog Evidence	41
7.6	Timing Loss and Recovery	42
7.7	Utility-Converter Synchronization Is Separate	42
7.8	Optional Board-Local Converter Synchronization	43
7.9	Timing Budgets Belong to the Application	43
8	Identity, Communication, and Data Architecture	44
8.1	Direct Host-to-Board Communication	44
8.2	Information and Configuration	44
8.2.1	Operational Configuration	44
8.2.2	Sequencer Transfer and Readiness	46
8.3	Service, Identity, and Board Replacement	46
8.3.1	Persistent Board Identity	46
8.3.2	Host-Owned Logical Inventory	47
8.3.3	Replacing a Board	47
8.4	Supervision and Science Traffic	48
8.4.1	Science Data Goes Directly to the Host	48
8.4.2	Data Overrun Is Not Automatically a Safety Fault	48
8.5	Trusted Instrument-Network Boundary	49
III	Applying the Architecture	50
9	Operating and Fault Scenarios	51
9.1	Normal Startup, Acquisition, and Disarm	51
9.2	Arm Attempted While One Board Is Not Ready	52
9.3	Three Different Power-Loss Cases	52
9.3.1	A common utility rail becomes invalid	53
9.3.2	One board loses local interlock power	53
9.3.3	The central eFuse removes complete system power	53
9.4	Digital Power Collapse Versus Frozen Logic	53
9.5	Distributed Sequencer-Clock Loss	54
9.6	Host Communication Loss During Acquisition	55
9.7	Data Overrun Without Supervision Timeout	55
9.8	Physical Discontinuity	55
9.9	Common Scenario Summary	56
9.10	Generic Fault Correction and Recovery	56
IV	Reference Appendices	58
A	Glossary	59
B	ADR Traceability Index	60
B.1	Handbook Chapter to ADR	60
B.2	ADR to Handbook Chapter	60

C	Shared-Signal Summary	61
D	Power-Rail Summary	62
D.1	System Input and Protected Bulk Rail	62
D.2	Mandatory Common Utility Rails	62
D.3	Board-Local Supplies	63
E	Official Project References	64
F	Version and Generation Information	65
F.1	Source Hierarchy	65

Part I

Context and Overview

1 Origins and Guiding Ideas

Astronomical instruments often operate for decades. Their detector controllers must remain useful through several generations of processors, data links, design tools, and electronic components.

A detector controller also has a critical role. It creates detector clocks and bias voltages, reads small video signals, and sends science data to the host. A controller failure can stop an instrument or place a valuable detector at risk.

The problem is therefore larger than replacing one obsolete part. A controller can become difficult to support because spare parts are gone, the original vendor no longer supports it, its host interface is outdated, its design files are unavailable, or only a few people know how it works. Replacing one weak part without improving the architecture may only move the problem to another part of the system.

This project addresses that long-term support problem with a modular detector-controller architecture. The project has already been approved. This chapter explains the ideas behind the architecture rather than repeating the project proposal or business case.

1.1 From a Local Need to a General Architecture

The project began with NOIRLab's need to support many detector controllers in long-lived instruments. An internal review found controllers of different ages, with different levels of vendor support, available spares, and local expertise. One important commercial controller family had lost vendor support. Existing in-house controllers were also affected by obsolete components and host interfaces. The detailed inventory and its dated assessments remain in the official project documents.

These problems are not unique to NOIRLab. Scientific instruments often remain in service much longer than commercial electronics. Detector systems are also built in small quantities, so a permanent commercial supply is unlikely. New detectors may need more channels or different combinations of video, clock, and bias functions. A direct replacement for one old controller would solve an immediate problem, but it would not provide a clear path for future instruments.

NOIRLab is the origin and first user of the project, but the architecture is intended to be open and useful to a wider detector-engineering community. It does not assume that one circuit can serve every detector. Instead, it defines common board roles and interfaces while allowing detector-specific electronics to remain specialized. The Phase II mandate moved the project from early concepts toward a controlled architecture, defined interfaces, and prototype testing.

1.2 Guiding Ideas

1.2.1 Plan for replacement

No electronic component will remain available forever. Long service life must therefore come from the ability to replace parts of the system without redesigning everything.

The architecture divides the controller into boards with clear roles and interfaces. A processor module, converter, or network device may change as technology changes. The replacement must still meet the same interface and system requirements. Open design files, test results, and recorded decisions allow future teams to understand and update the system.

This is what vendor independence means here. The controller still uses commercial parts. However, the instrument is not permanently tied to one supplier's private system architecture.

1.2.2 Share common functions and specialize detector electronics

Detectors do not all need the same analog circuits, voltages, timing patterns, or data rates. A single circuit designed to cover every case would become large and difficult to maintain.

The platform instead shares functions that are useful across many systems. These include board roles, power distribution, safety coordination, timing distribution, identity, service access, and network communication. Function boards provide the detector-specific video, clock, and bias circuits.

A board does not need to use every shared resource. It only needs to follow the common interfaces that apply to its role. This allows one detector interface to change without forcing unrelated boards to change with it.

1.2.3 Scale by adding modules

The architecture does not define one permanent controller size. More function boards, backplanes, or network capacity can be added when an instrument needs them. The main board coordinates the system, but it does not collect all science data or execute every detector function. This avoids making it the main data bottleneck.

Adding modules does not guarantee that every system size will work. Each instrument must still check its power, connectors, cooling, mechanics, network capacity, and timing. The architecture provides a consistent way to expand; it does not promise unlimited capacity.

1.2.4 Make safety and predictable behavior system properties

Safety must not depend only on host software or on one running processor. When hardware detects a safety-related fault, it drives the system to a safe condition. Software can then report and diagnose the cause. Shared interlocks, safe default outputs, local protection, and controlled recovery apply this rule across different function boards. ADR-001 and ADR-003 define this behavior.

The system must also behave in a predictable way. Timing-sensitive operations use controlled clock relationships. Management and protection continue to work if the acquisition clock is lost. Power conversion and input filtering are arranged so that switching noise can be measured and included in the system noise budget. The goal is not zero noise or zero timing error. The goal is behavior that engineers can analyze and test.

1.2.5 Separate architecture from implementation

The project uses different documents for different levels of detail. Architecture Decision Records define behavior and responsibilities that affect several boards. Interface Control Documents define the signals, messages, limits, timing values, and connector assignments at each boundary. Board specifications define how hardware, firmware, and software implement those interfaces.

This separation allows implementations to change without changing the architecture. It also prevents this handbook from becoming a second technical specification. If this handbook simplifies a mechanism, the resolved ADR remains authoritative.

1.3 Why This Approach Is Practical Now

Building a modular controller is more practical and cost-effective now than it was decades ago. Integrated processor and FPGA modules can provide software control, deterministic logic, memory, and standard networking in a compact form. Ethernet allows active boards to communicate directly with the host, without a proprietary host-interface card. Modern converters, data converters, timing devices, and protection components reduce the amount of custom support circuitry.

The design process has also improved. Multilayer PCB fabrication and assembly are more accessible. Modern design and simulation tools help teams develop complex boards. Version control, automated builds, and open collaboration help preserve both the design and the reasons behind it. These tools reduce development effort and improve reproducibility. They do not remove the need for careful analog design, timing analysis, safety analysis, and system testing.

The architecture therefore defines required capabilities, not preferred product families. It does not require a particular processor module, FPGA family, converter line, or design-tool brand.

1.4 Scope of This Handbook

The following chapters explain the controller from its external power, timing, and host connections to its detector-facing clock, bias, control, and video interfaces. They describe board roles, power and supervision, hardware safety, operating states, timing, identity, communication, science data, fault behavior, and system integration.

The handbook does not define connector pinouts, communication protocols, component-level circuits, FPGA implementations, or operating procedures. Those details belong in instrument requirements, ICDs, board specifications, and verification records.

Chapter 2 gives a short view of the complete system. Later chapters then explain each part in more detail.

Chapter sources: NOIRLab project report, Phase II mandate, and architecture overview (NOIRLab Detector Group 2025; NOIRLab-ES Detector Group 2025; NOIRLab Detector Group 2026). Current technical decisions are recorded in ADR-001 through ADR-006.

2 The System at a Glance

The controller is a set of boards connected through a backplane. Each board has a clear role. The boards share power, safety signals, and timing, but they do not depend on one central processor for all their work.

A remote host operates the complete instrument. It communicates directly with every active board over Ethernet. Function boards perform detector-specific work. The main board coordinates the system. The backplane provides the common physical and electrical infrastructure.

Figure 2.1 shows this arrangement. It is a simple system view, not a wiring diagram. Later chapters explain each connection in more detail.

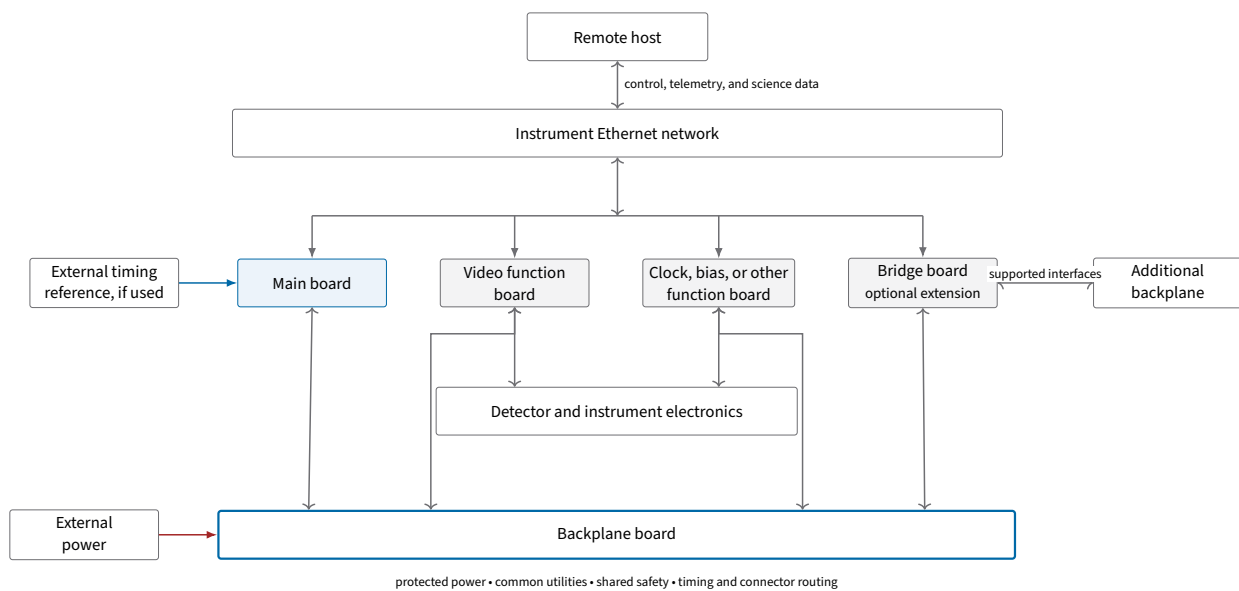


Figure 2.1: High-level controller structure. Science data travels directly between the host and the function board that produces it; it does not pass through the main board.

2.1 Board Roles

The architecture uses five board roles. A controller may contain several function boards and may use a bridge when it needs more than one backplane.

Board role	Main purpose
Main board	Coordinates system-wide enable, recovery, timing, and synchronization. It also observes shared health. It does not collect all science data or control the internal work of every function board.
Function board	Performs detector-specific work such as video acquisition, clock generation, or bias generation. Each active function board has its own control and network endpoint.
Backplane board	Holds the slots and distributes protected power, common utility rails, shared safety signals, timing, and connector routing.
Bridge board	Extends the supported shared interfaces to another backplane when the system must grow.
Passive terminator	Completes the safety continuity path in an unused slot. It has no active electronics.

The main board coordinates the controller, but it is not the controller's only intelligent element. Each active board manages its own local function, reports its status, and responds to the shared safety rules. This keeps the main board from becoming a processing or data bottleneck.

2.2 The Four Main Connections

Four types of connection make the separate boards operate as one controller.

2.2.1 Power

External power enters the backplane and passes through central input protection. The protected bulk supply then feeds every active board and the backplane utility converters. Common analog utility rails are available to boards that need them. Each board generates its ordinary digital supplies locally from the protected bulk supply.

2.2.2 Safety and coordination

The boards share a hardware fault path. Any participating board or supervised common rail can place the controller in a safe condition. The main board also provides the common arm and recovery signals. Detector-facing outputs remain off until both the system and the local board allow them to operate.

Software is used to identify and report the cause after the hardware has made the system safe. A lost host connection, failed processor, or missing acquisition clock must not prevent the required safety response.

2.2.3 Timing

The main board distributes the clock and synchronization events used for detector acquisition. Function boards use these signals when their work must align with other boards. Each active board also has an independent local clock for management, communication, and fault monitoring. Losing the acquisition clock therefore stops synchronized work without removing the board's ability to report the fault.

2.2.4 Communication and science data

Every active board has a direct Ethernet connection to the host. The host keeps the instrument inventory, maps each logical role to the correct board, sends configuration, and reads status.

Science data follows the same direct principle. A video board sends its data to the host instead of routing it through the main board or through a shared backplane data concentrator. This allows data capacity to grow as video boards and network capacity are added.

2.3 A Normal Operating Cycle

The complete controller follows a simple operating story:

1. **Start safely.** Boards power up with detector-facing outputs disabled. They check local power, logic, timing, communication, and shared health.
2. **Configure the instrument.** The host identifies the active boards and sends the settings and sequencer data needed for the current session.
3. **Arm deliberately.** The host checks readiness and requests arming. Main checks its own conditions and raises EN. Each function board independently decides whether it can accept the arm request.
4. **Acquire data.** The main board sends synchronized events. Function boards execute their local timing and data tasks. Video boards send science data directly to the host.
5. **Disarm or go safe.** Stopping acquisition may leave the controller armed. Disarming drops EN and removes detector-facing permission. A trip removes permission through the interlock and requires explicit recovery.
6. **Recover explicitly.** Restoring a missing clock, rail, or network link does not restart acquisition. The host requests recovery, the boards repeat their checks, and the system must be armed again.

Chapter 6 follows these actions through the operating states.

Chapter sources: ADR-001 through ADR-006 and the repository system concept guide. The ADRs remain authoritative for technical behavior.

Part II

Architecture

3 Modular System Structure

The controller is divided into parts that can be designed, tested, and replaced separately. This is the main purpose of the modular structure. It allows the system to change over time without turning every upgrade into a complete controller redesign.

Modularity depends on clear boundaries. A removable board is not truly independent if changing it also requires undocumented changes in several other boards. The architecture therefore assigns each system responsibility to an owner and places shared behavior in controlled interfaces.

Figure 3.1 gives a simple physical picture of the modular idea. A main board and specialized function boards plug into a common backplane. The drawing is conceptual: the number, size, and arrangement of boards depend on the instrument.

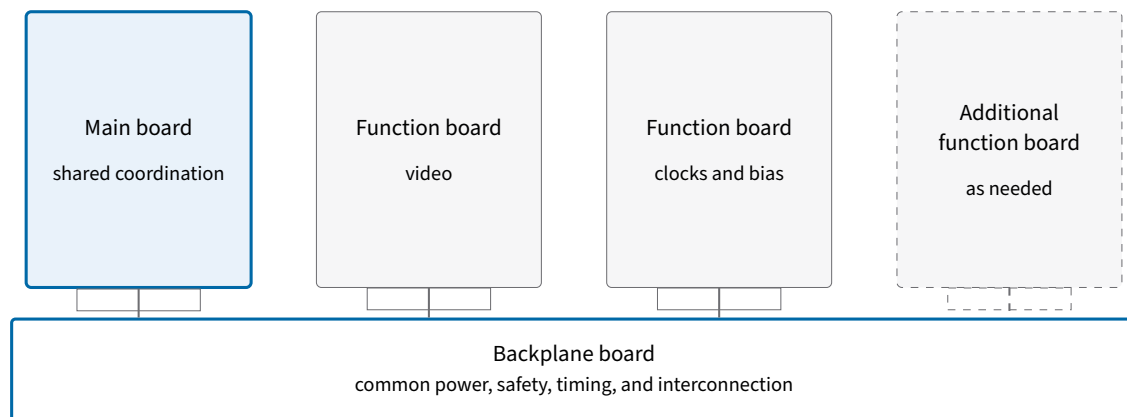


Figure 3.1: Conceptual modular arrangement. The backplane provides the common platform, while removable boards provide coordination or detector-specific functions.

Figure 2.1 shows how these physical elements connect to the wider instrument. This chapter explains the responsibilities behind the structure.

3.1 A Stable Platform with Specialized Function Boards

The architecture separates common platform services from detector-specific electronics.

The common platform provides protected power, utility rails, safety coordination, timing distribution, communication access, and mechanical connections. These services are useful to many detector systems and should behave consistently.

Function boards provide the electronics that depend on the detector. A system may use video boards, clock boards, bias boards, or a board that combines several functions. A new detector may require a new function board without requiring a new main board or a new backplane architecture.

This separation does not mean that every function board is simple or identical. It means that each board connects to the rest of the controller through a known set of contracts. A board uses only the shared

resources it needs and provides the safety, timing, communication, and electrical behavior required for its role.

3.2 Main Board

The main board coordinates system-wide actions. It generates or distributes the signals that must be common across several function boards.

Its main responsibilities are:

- driving the shared arm and recovery signals;
- generating the acquisition clock and synchronization events;
- providing synchronization signals for the backplane utility converters;
- monitoring the shared fault bus and continuity loop;
- measuring common power rails for diagnosis; and
- communicating with the host as its own active board.

Detector sequencers, detector-specific electronics, video processing, and science-data transport belong to function boards. Main coordinates shared actions.

This limited role is important for scaling. Replacing a video board with a faster design should not require a faster main-board data path. Adding a detector-specific function should not require adding that function to the main board.

The main board can coordinate shared electrical behavior without knowing which logical instrument role is installed in each slot. The host owns that inventory. This keeps physical position separate from operational identity.

3.3 Function Boards

A function board performs work that is specific to a detector or instrument. Typical examples include:

- digitizing detector video;
- generating detector clocks;
- generating detector bias voltages;
- controlling detector support electronics; or
- combining several related functions when an instrument needs it.

Each active function board contains the processing, firmware, local conversion, monitoring, and network connection needed for its own work. It receives common timing and coordination signals from the platform, but it decides locally whether it is ready to operate. It also participates directly in the shared hardware safety path.

A function board does not command another function board. Coordination comes from the host and the shared signals driven by the main board. This avoids hidden dependencies between particular board types.

Function boards are also responsible for their detector interfaces. Their ICDs define voltage ranges, timing, loading, data capacity, and other limits at those interfaces. Specialized detector voltages remain local to the board that needs them. Common rails are used only when they fit the function and noise requirements.

The host configures each function board and reads its diagnostics directly. Chapter 8 explains communication and science-data transport.

3.4 Example: Inside a Function Board

A function board is a removable PCB assembly connecting the common controller infrastructure to a detector function. Figure 3.2 uses a video board to show what that means. The outline and zones explain the parts and their relationships; they are not a PCB placement plan.

The backplane connector carries fixed power contacts, timing and coordination signals, OK, returns, and continuity-loop contacts. Local conversion supplies ordinary digital loads from protected +12V. Analog loads may use the shared utility rails, local conversion, or both.

Processing and memory configure the function, capture or process samples, and handle host traffic. Timing hardware receives acquisition events, while local management and safety clocking remains independent where required. Interlock hardware includes the watchdog, fail-safe paths, supply supervision, and an optional dedicated safety CPLD. It controls applicable detector-facing permissives independently of ordinary processing.

The functional electronics in this example comprise input protection, analog conditioning, an ADC, and local monitoring. The detector interface connector links them to the detector or preamplifier. Ethernet connects directly to the host; UART provides local commissioning and recovery. The continuity path may use passive copper or unconditional hardware forwarding in the safety CPLD. It is independent of software and operating state.

A clock or bias board replaces the video chain with drivers or DAC/output stages. The common interfaces retain the same meaning. A local voltage/current monitor does not replace analog input protection, and the illustration does not prescribe a relay circuit or processor family.

3.5 Backplane Board

The backplane is more than a passive collection of connectors. It provides the common physical and power infrastructure for the controller.

Its responsibilities include:

- protecting the external power input and distributing protected bulk power;
- generating the standard utility rails;
- supervising the shared rails and contributing faults directly to the shared safety bus;
- routing timing, coordination, safety, and continuity signals;
- providing the electrical and mechanical slot interface; and

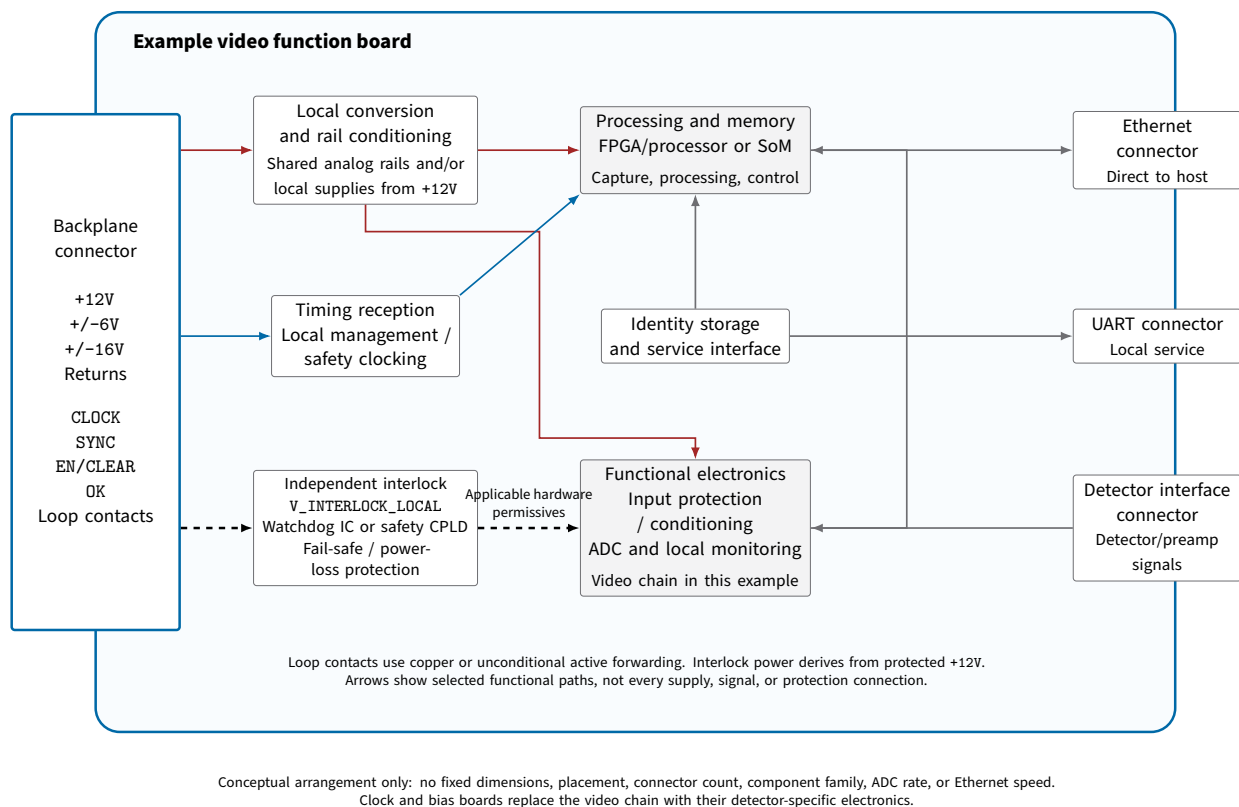


Figure 3.2: Conceptual video function board with its connectors and main functional parts. Arrows show selected relationships, not complete wiring. Dimensions, component choices, connector geometry, sampling rate, and link speed remain application-specific.

- maintaining deliberate analog, digital, return, and shielding paths.

The utility converters live on the backplane because their rails are common platform resources. Boards that need high-current digital power create it locally from protected bulk power. This avoids sending large processor and FPGA currents through low-voltage shared-rail contacts.

The backplane does not interpret instrument commands or control detector operation. Its active circuits provide power, protection, supervision, and signal distribution. Detailed thresholds, connector assignments, current limits, and routing rules belong to the backplane ICD and hardware design.

Every standard backplane presents the same set of common rails, even when a particular function board does not use all of them. Removing or changing a standard rail creates an architecture variant rather than a normal board option. This single definition keeps board and backplane documentation manageable.

3.6 Bridge Boards and Additional Backplanes

A bridge board allows the controller to use another backplane when one backplane is not enough. It carries the supported shared interfaces across the extension and reproduces active signals where required.

The extension must preserve the meaning of the original interfaces. The shared fault bus remains one hardware interlock. The continuity loop must travel through the extension and return to the main board, so a disconnected cable is detected. Timing must still reach each supported slot through a verified point-to-point path.

A bridge board is an active board and therefore has its own communication, health monitoring, watchdog, and local interlock power. Its continuity path may use passive copper or qualified active forwarding. A bridge power failure trips through a separate local hardware contribution or through active forwarding and main-board conversion, respectively.

The architecture does not set one universal number of slots or backplanes. Each instrument selects a topology and verifies the complete power, timing, network, thermal, and mechanical budgets. An available connector does not prove that the system has capacity for another board.

3.7 Passive Terminators and Unused Slots

The continuity loop passes through every slot. An unused slot therefore needs a passive terminator to complete the loop. The terminator is a small unpowered insert with the required copper connection. It is not an active board and has no identity, network connection, or local interlock supply.

Removing a board or terminator breaks the loop and places the system in a safe condition. Live insertion and removal are not normal operating modes. Board replacement is performed with the controller powered down or already safely latched.

3.8 Responsibility Boundaries

Table 3.1 summarizes the main ownership rules. The goal is not to list every circuit. It is to show where a design change should begin and which interface must remain stable.

Owner	Owns	Does not own
Remote host	Instrument inventory, configuration, operation, and data reception	Hardware interlock action or board-local safety decisions
Main board	Shared enable, recovery, acquisition timing, synchronization, and continuity-loop monitoring	Detector sequencers, detector-specific electronics, or science-data concentration
Function board	Its detector function, local readiness, local protection, diagnostics, and direct host traffic	System-wide signal generation or control of other function boards
Backplane board	Protected power, common utility rails, shared-rail supervision, slots, and common routing	Instrument commands or detector-specific behavior
Bridge board	Verified extension of supported interfaces to another backplane	Redefinition of those interfaces or independent system coordination
Passive terminator	Continuity through an unused slot	Any powered or programmable function

Table 3.1: High-level ownership boundaries. Exact interfaces and limits are defined in the applicable ICDs.

These boundaries make common changes easier to contain:

- An obsolete processor can be replaced inside a board design if the board still meets its interfaces.
- A new detector can use a new function board while reusing the common platform.
- More channels can be added by composition after the complete system budgets are checked.
- A change to a shared signal, rail, or board role is an architecture or ICD change because it affects several independent elements.

System integration verifies that the installed boards meet their declared interface and capacity limits.

Chapter sources: ADR-001 through ADR-006, with the main ownership rules in ADR-002 through ADR-005. The repository README defines the document and terminology boundaries.

4 Power, Supervision, and Noise Architecture

Every standard backplane provides the same power interface. Each function board chooses which supplies to consume and which to generate locally. Shared rails and local conversion are combined according to the loads, noise limits, and thermal budget.

Power supervision follows the same division of responsibility. The backplane protects and supervises shared power. Each active board protects its own safety-related functions. The main board measures common rails to help the host diagnose a trip.

Figure 4.1 shows the complete power path and the main supervision paths.

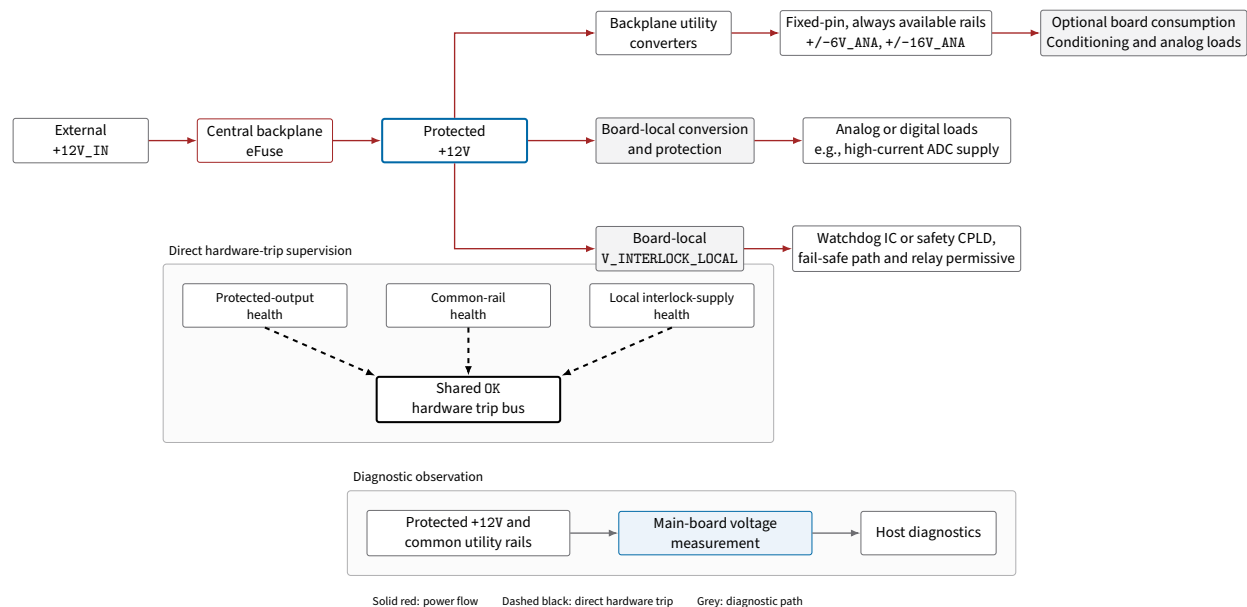


Figure 4.1: Power and supervision overview. Red lines show power flow, dashed black lines show direct hardware contributions to the shared fault bus, and grey lines show diagnostic measurement.

4.1 What Power Is Available?

4.1.1 Protected bulk power

The external system input is called +12V_IN. It enters the backplane and passes through one central eFuse. The eFuse output is the protected rail called +12V.

The two names identify an important protection boundary:

+12V_IN → central backplane eFuse → protected +12V

The central eFuse acts as the system input circuit breaker. It protects against unsafe input voltage, reverse polarity, excessive total current, and system inrush. It also prevents the protected +12V rail from

remaining outside the verified input range of downstream converters. If that range cannot be maintained, it disconnects all downstream power. The utility converters and every active board are behind this device.

The exact eFuse part, limits, delays, and restart behavior are backplane design choices. They are selected from the system power requirements and verified operating conditions.

4.1.2 Common analog rails

Shared analog utility supplies let boards reuse common conversion where it suits their loads. Ordinary digital supplies remain locally generated from protected +12V.

Every standard backplane provides the following utility rails on fixed connector assignments.

Rail	Intended use
+6V_ANA	Common positive low-voltage analog utility rail
-6V_ANA	Common negative low-voltage analog utility rail
+16V_ANA	Common positive analog utility rail
-16V_ANA	Common negative analog utility rail

Table 4.1: Standard backplane utility rails, generated from ADR-005.

A board does not have to consume every utility rail. An analog board may use several of them, while a specialized board may use only protected +12V. The standard interface remains the same in both cases. A backplane that removes or repurposes a standard rail is a separate architecture variant and needs an explicit decision and ICD.

4.2 How a Board Uses the Available Power

Function boards may use any subset of these rails, leave unused rail contacts unconnected, or generate supplies locally from protected +12V, even when an equivalent nominal voltage is available on the backplane. Local conversion is a normal application-dependent choice, not an architecture variant. A high-current ADC supply may use local conversion while other analog loads use shared rails. Voltage, current, noise, transient response, efficiency, and thermal requirements guide this choice.

Ordinary digital supplies are different. Each board generates its processor, FPGA, memory, identification, monitoring, and management supplies locally from protected +12V. The standard backplane does not distribute a low-voltage digital utility rail because no common use case justifies its converter, connector allocation, supervision, and shared-noise path.

4.2.1 Local conversion and protection

Processors, FPGAs, and memory can draw substantial current at low voltage. Distributing that current directly at 3.3 V would require more connector contacts and heavier copper. It would also increase voltage

drop and expose a common rail to fast digital load changes from every board.

The architecture therefore sends protected +12V to each active board. The board converts it locally to the voltages needed by its processor, FPGA, memory, and other high-current loads. Higher distribution voltage keeps the shared current lower for the same delivered power. Local conversion also keeps fast point-of-load behavior close to the circuits that create it.

The same +12V rail is the normal source for uncommon detector voltages. A specialized bias board may need voltages that are not useful to the rest of the controller. Those converters remain on that board instead of adding uncommon rails to every backplane slot.

Each board is responsible for the loads behind its connector. Its ICD declares steady current, peak demand, inrush or input capacitance, voltage tolerance, and sequencing needs for every rail it consumes.

Both main and function boards may use optional input eFuses on protected +12V or any consumed analog utility rail, using protection circuitry suitable for the rail's voltage and polarity. A fuse, load switch, protected converter, or another suitable method may also provide local protection. The choice depends on the board's energy, failure, and service risks. What matters is that a board fault must not damage the board, connector, or shared distribution.

Common analog rails usually feed filters and LDOs at the board boundary. These circuits provide regulation and noise control, but they are not automatically circuit breakers. The architecture does not require a separate eFuse on every analog-rail input.

The architecture also does not require independent supervision of every local converter. A local monitor is added when failure of that rail or output could leave a safety-related function in a hazardous condition that is not already covered by the board's fail-safe and watchdog paths.

4.2.2 Conditioning shared rails

Every shared power rail connects the noise behavior of the boards that use it. This includes protected +12V as well as the shared analog utility rails. Incoming noise can disturb local circuitry, while a board can return load transients and conducted emissions to the rail and disturb another board.

Digital loads and board-local converters must also control the disturbances they inject into protected +12V. Another board may use that same bulk rail to generate sensitive analog supplies. Local conversion does not remove the shared input-noise path.

Each function board therefore conditions every consumed shared rail at its power-entry boundary. As shown in Figure 4.2, the conditioning has two jobs:

1. reduce incoming rail disturbances enough for the local circuitry; and
2. limit disturbances returned by the board to the common rail.

The board design may use an LDO, passive filter, ferrite and local storage, active filter, or another verified circuit. The architecture does not choose one topology or set component values. The backplane ICD defines the rail environment, and each function-board ICD declares the rejection it needs and the disturbance it is allowed to return.

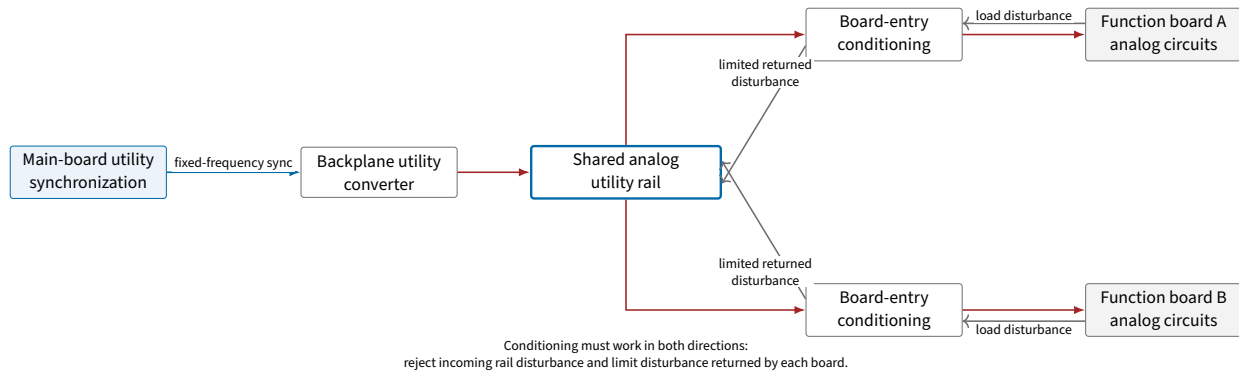


Figure 4.2: Shared analog-rail example of bidirectional conditioning. The same incoming-noise and returned-disturbance requirements apply to protected +12V, including digital loads and local converters. Utility-converter synchronization makes the switching pattern predictable; conditioning does not imply zero noise.

4.2.3 Returns, connectors, and budgets

Power integrity also depends on return-current paths and connector allocation. The architecture does not require separate physical connectors for analog and digital signals. One connector can be used if its pinout and routing keep analog power, digital infrastructure, fast signals, and their returns in deliberate zones.

Analog-power and digital/infrastructure returns use dedicated contacts. Fast differential pairs have nearby returns. Analog utility contacts are separated from fast digital signals by return contacts or an equivalent grounded boundary. Chassis and shield connections remain distinct from circuit returns. The backplane design defines where analog and digital circuit returns are intentionally related.

Connector current capacity is not only a steady-state calculation. The design must also check contact resistance, voltage drop, temperature rise, inrush, peak demand, copper heating, and reliability. Parallel contacts are used where the verified current requires them.

The same rule applies to the complete controller. An empty physical slot does not prove that enough power, cooling, timing, or network capacity remains for another board. Integration checks the worst-case installed set of boards against the per-slot and total budgets.

4.3 What Happens When Power Fails?

4.3.1 Complete input cutoff

The eFuse is independent of the normal arm and fault state. A shared interlock trip makes detector-facing functions safe, but it does not normally turn off the complete controller. This allows boards to retain diagnostics and communicate with the host. The eFuse removes system power only when required by its own input-protection function.

If the eFuse disconnects the complete system, all active boards and powered outputs turn off. Normally-open detector relays also de-energize. The architecture does not require the shared fault bus or telemetry to remain active after this complete loss of power.

4.3.2 Supervision and diagnosis

Power-good signals, voltage measurements, fail-safe circuits, and watchdogs are easy to confuse because they may all lead to a system trip. Their detection roles remain distinct even when some functions share a device.

Mechanism	What it detects	Role in the system
Central 12 V protection	Unsafe +12V_IN, invalid protected +12V, total overcurrent, reverse polarity, or inrush	Pulls OK low when possible and disconnects the complete downstream power tree when required
Backplane utility-rail health	Invalid common utility rail	Pulls the shared OK bus low directly; it does not command complete 12 V cutoff
Main-board measurement	Actual common-rail voltages and optional currents	Provides telemetry and helps the host identify the cause; it is not the fast protection path
Interlock-power-loss reporting	Loss of one board's independent interlock supply	Causes a shared trip directly or through qualified active forwarding and main while the fleet remains powered
Fail-safe power/reset path	Loss or reset of processor or FPGA power while interlock power remains valid	Pulls OK low without depending on the failed logic
Independent hardware watchdog	Missing logic activity or a failed required pet path	Pulls OK low as a liveness fault; it is not a voltage monitor
Board-specific monitor	A local safety-related voltage or output not covered by the common paths	Enters that board's normal local-fault path when the board design requires it

Table 4.2: The power, diagnostic, and liveness mechanisms have different scopes.

For a common utility converter, its native PGOOD output may be used as the rail-health signal if it covers the required voltage conditions and is electrically compatible with the shared OK bus. If it does not, the backplane adds a simple supervisor or interface. A second monitor is not required when the native output already provides the needed coverage.

In this architecture, *PGOOD* means the native power-good output of a converter. The term is not used for a watchdog, a fail-safe output, or the supervisor for local interlock power.

The qualified rail-health outputs connect directly to the shared OK bus. Individual PGOOD copies are not sent to the main board. This avoids adding a separate buffer or isolation channel for every rail. After a trip, the main board reads the rail voltages and the host polls board diagnostics to identify the likely cause. A very short event may remain identified only as a common-power trip unless the backplane stores additional evidence.

4.3.3 Local interlock power

Every active board derives one small supply called `V_INTERLOCK_LOCAL` from protected +12V. This supply powers only the independent protection circuits: the independent hardware watchdog, fail-safe fault driver, relay-permissive hardware, and power-loss protection hardware. A dedicated safety CPLD may also use this supply for the interlock functions permitted by ADR-001.

It does not power the processor, acquisition FPGA, memory, analog signal chain, or detector load. It is also not a backplane utility rail. The word *local* means that each active board creates its own interlock supply.

If one board loses this supply while the rest of the system remains powered, its hardware power-loss reporting must cause OK low long enough to trip the fleet, either directly or through qualified active loop forwarding and main-board conversion. Loss of interlock power must also disable detector protection relay permission, even if relay-driver power remains available. A dedicated supervisor is optional: the board design must verify both local relay disable and the F6 shared trip. Stored energy is one possible implementation choice.

This case is different from a complete central eFuse cutoff. When the complete controller loses power, all hazardous powered outputs de-energize directly. There is no need to keep one unpowered board reporting to an unpowered fault bus.

4.3.4 Startup and common-rail faults

The mandatory utility converters remain enabled whenever protected +12V is available. They are not turned off by EN, OK, or the system state. This avoids a circular recovery problem in which a fault removes a required rail and its PGOOD can never return to the healthy state.

During startup, rail-health outputs may hold OK low until the common rails become valid. The system cannot arm during this period. If a rail never becomes valid, startup fails and the controller remains safe.

During operation, loss of a common rail pulls OK low and sends the controller to its safe error state. If the voltage returns, acquisition does not restart. The system remains latched safe until the host requests recovery and the live conditions pass qualification again.

4.4 Predictable Converter Switching

Switching converters create ripple and conducted emissions. Their noise cannot be removed by synchronization alone, but their behavior can be made more predictable.

During acquisition, every enabled backplane utility converter uses a synchronization signal and operates in forced continuous, fixed-frequency mode. Burst mode, pulse skipping, automatic low-frequency operation, and spread-spectrum operation are not used for these converters. Their switching pattern therefore remains stable enough to measure and include in detector noise analysis.

The main board provides five point-to-point LVDS synchronization outputs for the backplane utility converters. The lines reserve support for up to five switching channels, but they are not permanently assigned one-to-one to the utility rails. A converter design may use fewer switching channels or combine rails.

Each synchronization frequency is an integer division of the common 2 MHz baseline. The backplane ICD defines the channel mapping, divisor, phase behavior, capture range, and electrical details. Those settings remain fixed during acquisition.

Before synchronization is present, or if it is lost, an enabled converter continues in a defined fixed-frequency, forced-continuous free-running mode. Synchronization loss must not create an uncontrolled rail or require processor action. Utility-converter synchronization is separate from the 100 MHz detector sequencer clock and acquisition SYNC described in Chapter 7.

Compatible converters on a function board may optionally derive their switching references from that board's distributed timing. Whenever the system enters START, main sends one alignment SYNC pulse while EN=0; compatible boards recognize any such pulse and align their local divider before reporting ready. With EN=1, acquisition SYNC events do not alter the divider. This board-local option is separate from the five dedicated backplane utility-converter synchronization outputs.

Chapter sources: ADR-005 defines bulk power, utility rails, shared-rail supervision, synchronization, conditioning, and connector boundaries. ADR-001 defines local interlock power and independent protection paths. ADR-003 defines startup, fault entry, and recovery behavior.

5 Hardware Safety and Interlock Architecture

The safety architecture follows one rule: a detected safety-related fault makes the controller safe through hardware. Software identifies and explains the cause afterward.

This order matters because the host, processor, FPGA, or acquisition clock may be part of the failure. None of them can be the only path that removes a hazardous detector-facing output.

The architecture starts from outputs that are safe when unpowered. It then combines several independent detection paths into one shared interlock. A fault on one board removes permission from every affected function board.

5.1 Start at the Detector-Facing Output

Detector protection relays are normally open. They require power and permission to energize. Loss of board power therefore opens the relay without waiting for software or firmware.

While power is available, a relay may energize only when these common conditions are true:

$$\text{relay_energized} = \text{local_arm_request} \text{ AND } \text{EN} \text{ AND } \text{OK}$$

The three conditions come from different parts of the system:

- `local_arm_request` means the function board has completed its own readiness checks and deliberately requests operation;
- `EN` means the main board has armed the controller; and
- `OK` means no participant is currently asserting the shared hardware fault bus.

A board using active forwarding may add incoming-loop validity as another hardware permissive. Incoming LOOP LOW then removes local detector-facing permission without waiting for main, as described in Section 5.4.

Figure 5.1 shows the common permission path. The exact gate, latch, relay driver, and relay part are board-design choices. The required behavior is that loss of `EN`, `OK`, or local interlock power removes relay drive without depending on processor software, FPGA/CPLD state progress, or the distributed acquisition clock.

Arming is therefore deliberate, while going safe is direct. A board cannot keep a detector protection relay energized merely because its FPGA still believes that it is in a running state.

5.2 Three Safety Layers

The complete system uses three cooperating layers. They have different response times and different dependencies.

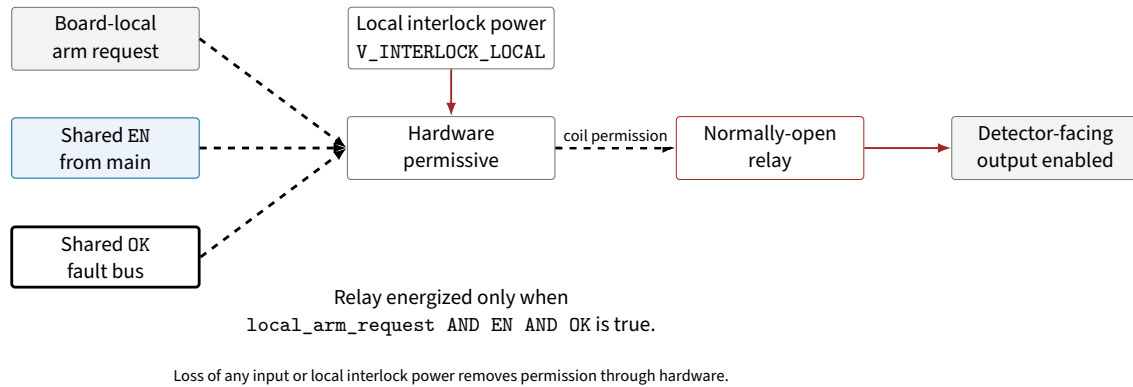


Figure 5.1: Common detector-facing hardware permissives; optional incoming-loop qualification is not shown. All required permissions must remain valid to energize a normally-open relay.

Layer	Main job	Important boundary
Hardware interlock	Removes detector-facing permission when a shared or local trip occurs	Does not wait for host software or continued FPGA/CPLD operation
Board safety FSM	Checks readiness, retains fault evidence, controls local state, and coordinates recovery	Uses independent local management or safety clocking; FPGA or safety CPLD logic cannot override a hardware trip
Host management	Configures boards, coordinates instrument operation, and gathers diagnostics	Can request arm or recovery but cannot force hardware permission while a trip is active

Table 5.1: The safety layers cooperate without giving software authority over the final hardware interlock.

The layers support one another. Hardware provides the fast and independent safe response. The board FSM keeps enough state for orderly operation and recovery. The host gives the operator a complete instrument view. Removing any one layer would either weaken protection or make the system difficult to operate and diagnose.

5.3 The Shared Open-Drain Fault Bus

The OK signal is a shared open-drain bus. The main board provides one pull-up. Every participating board and qualified backplane rail-health source may pull the bus low, but no function board drives it high.

Its meaning is intentionally simple:

- OK=1: no participant is currently asserting the bus;
- OK=0: at least one participant is asserting a trip.

Because the bus is wired together, one fault removes permission across the controller. A board cannot mask another board's low assertion by driving high. The bus also remains useful when several sources trip at the same time.

The OK level does not identify the source. A healthy board enters its safe response when it sees the bus low, but it does not claim that it caused the trip. A board pulls the bus low only for its own retained local fault or supervisory event. After the safe response, the host reads diagnostics from the boards and common-rail measurements to find the cause.

This distinction prevents a fleet-wide trip from becoming a permanent loop in which every healthy board asserts its own fault output only because it observed OK low.

5.4 The Continuity Loop

The open-drain bus detects electronic faults, but it cannot by itself prove that every connector and cable remains present. A removed board cannot pull OK low. The architecture therefore adds a continuity loop.

The main board sends LOOP_OUT through every slot and receives the returning signal as LOOP_IN. Each board passes the path through passive copper or unconditional combinational hardware forwarding. Both types may coexist under the common electrical interface. It also passes through bridge cables and secondary backplanes when they are present.

If a board, passive terminator, connector, or extension cable opens the path, LOOP_IN becomes invalid. Deterministic hardware on the main board converts that condition into its own OK assertion. Processor software is not the conversion path.

An unused slot contains a passive terminator that completes the loop. Removing an active board or terminator breaks the path and trips the system. This is intentional. Operational hot-swap is not supported.

During normal operation, active forwarding copies the incoming logic level independently of OK, EN, operating state, and clocks. On interlock-power loss it establishes LOW toward the next receiver. An

output verified to become high impedance with an external pull-down is one implementation. The board may also make its outgoing loop LOW to report a detected condition that prevents its safety hardware from performing its role. Forwarding resumes when power and configuration are valid and that condition clears, without waiting for the retained trip to clear. Coverage of every possible internal device failure is not required. The complete chain must be valid before arming.

Implementation	Continuity	Interlock-power loss (F6)
Passive copper	Direct connection	Separate hardware contribution to OK required
Active forwarding	Unconditional copy	Qualified LOW default propagates to main, which asserts OK

Table 5.2: Permitted board continuity implementations. Empty-slot terminators remain passive.

A function board using active forwarding may use incoming LOOP_IN LOW to remove local detector-facing permission directly in hardware. This fast local response does not wait for main or a state-machine transition. LOW continues downstream to main, which asserts OK so every powered board, including those upstream of the fault, enters the shared safe response.

Local safe-state entry must not itself force the outgoing loop LOW. Forwarding remains independent of OK, EN, and operating state; it copies the input unless local power or a detected safety-hardware condition requires LOW. This avoids a recovery feedback loop. The chain propagates an upstream problem but does not automatically identify the affected board.

The backplane ICD defines loop voltage and logic thresholds, drive strength, leakage and loading, pull-down placement and values, unpowered and partial-power behavior, and total propagation time. Verification includes consecutive passive boards and terminators, configuration behavior, and prevention of back-powering. The architecture does not select a loop voltage.

Qualified active forwarding can provide F6 reporting without a separate local supervisor or hold-up supply. It does not prove that the rest of the CPLD is executing correctly and does not replace the watchdog. Main's own power-loss response needs a verified path independent of its failed conversion logic. A LOW return alone does not distinguish physical discontinuity (F1) from power loss in a forwarder (F6).

The continuity loop and OK bus solve different problems. The loop detects physical interruption and may also report interlock-power loss through qualified active forwarding. The open-drain bus carries electronic, timing, power, liveness, and supervisory trips. Both are required.

5.5 Independent Fault Paths

No single monitor can detect every important failure. The system uses several simple paths that converge on the shared OK bus. Figure 5.2 shows that convergence, and Table 5.3 summarizes the main cases.

The table does not mean that every local voltage needs a separate monitor. Additional local supervision is required only when an uncovered failure could leave a hazardous function active. Board hazard analysis determines those cases.

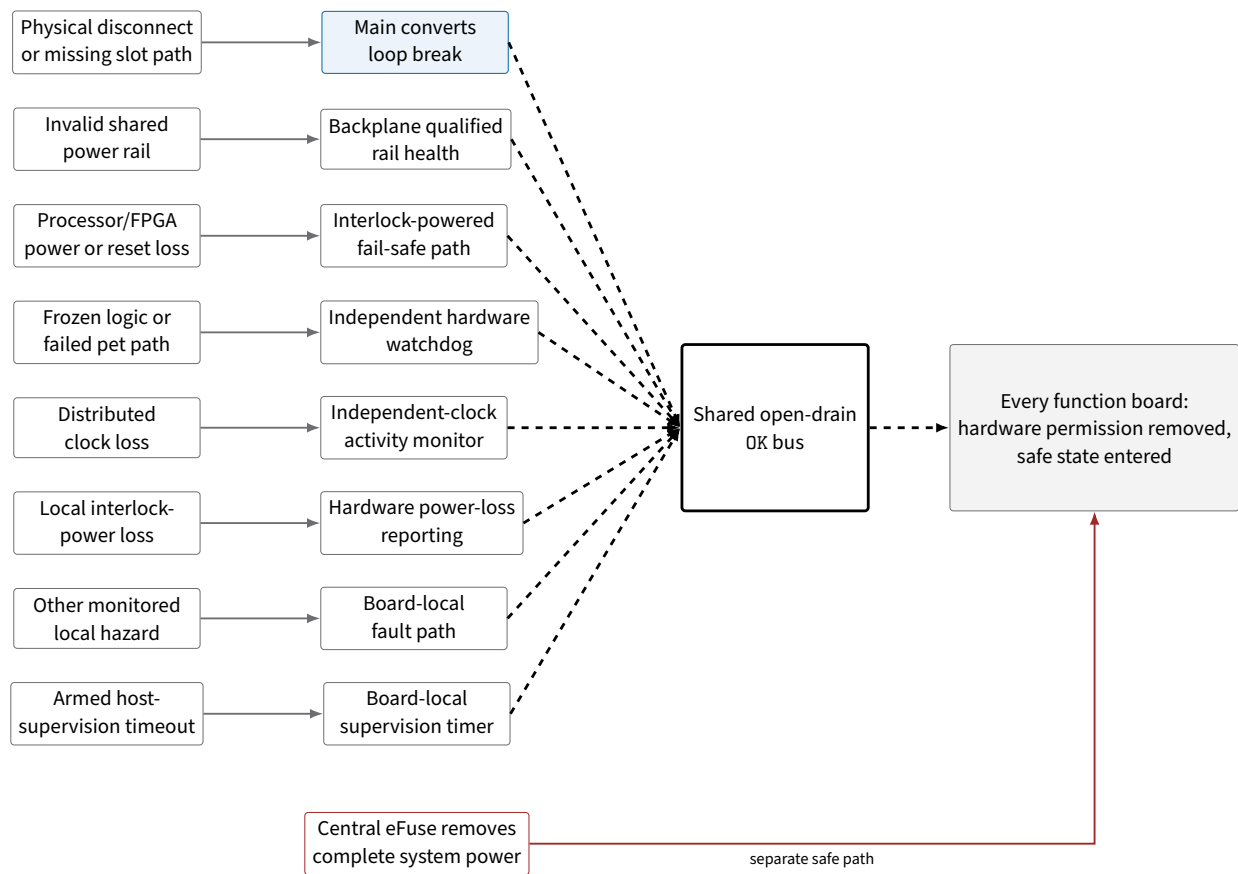


Figure 5.2: Independent fault sources converge on the shared hardware trip and the same fleet-wide safe response. Complete central power removal is a separate safe path.

Failure or event	Primary detection path	Safe propagation
Physical disconnect or open slot path	Continuity loop, observed by main	Main pulls OK low
Invalid protected +12V or utility rail	Qualified backplane rail-health output	Backplane pulls OK low directly
Processor or FPGA rail loss or reset	Interlock-powered fail-safe circuit	Local hardware pulls OK low
Frozen logic or failed required pet path	Independent hardware watchdog	Watchdog path pulls OK low
Distributed acquisition-clock loss	Activity monitor running from an independent local clock	Board-local fault path pulls OK low
Loss of V_INTERLOCK_LOCAL on one board	Hardware power-loss reporting path	Hardware asserts the shared trip; local relay permission is removed
Other safety-related local condition	Board-specific monitor, where the hazard requires it	Board-local fault path pulls OK low
Loss of host supervision while armed	Board-local supervision timer	Timed-out board uses its normal local trip path
Complete central eFuse cutoff	Central input protection	All powered outputs and normally-open relays de-energize directly

Table 5.3: Simplified mapping from representative failures to their protection paths.

5.6 Protection Must Survive the Failure It Detects

The element that fails cannot also be the only means of detecting its failure.

5.6.1 Power independent of the processor rails

Each active board creates V_INTERLOCK_LOCAL from protected +12V. It powers the watchdog, fail-safe output, relay-permissive hardware, power-loss protection hardware, and any dedicated safety CPLD. It is separate from ordinary processor and acquisition-FPGA supplies.

If processor or FPGA power disappears, the interlock-powered fail-safe path treats loss of the active health signal as unsafe and asserts OK. If interlock power itself fails, its hardware power-loss path trips the powered fleet directly through OK or through qualified active loop forwarding and main, and the local relay releases. Complete system-power removal instead de-energizes powered hazards directly.

5.6.2 A watchdog independent of the supervised clocks

A powered but frozen processor or FPGA may leave a stale healthy output. The watchdog detects missing progress through its service signal, often called *petting*. The processor or control logic refreshes it during normal execution; an autonomous toggle that continues after execution freezes is insufficient. Refresh

timing uses local clocking and may optionally align to acquisition timing for noise control, with local fallback when that reference is absent.

The watchdog timeout time base remains operational when supervised clocks stop. A watchdog inside the supervised processor or FPGA may add coverage, but cannot replace this independent path. Clock-activity monitoring likewise uses independent local management or safety clocking.

5.6.3 Implementing the functions with a safety CPLD

A dedicated watchdog IC or a dedicated safety CPLD may implement the watchdog. A safety CPLD may also combine fault aggregation and retention, clock monitoring, main-board continuity-loop conversion, relay permissive logic, and evaluation of voltage/current-monitor indications. Analog sensing still needs suitable circuitry. The hardware F6 reporting path must remain effective when interlock power fails; a dedicated supply supervisor is optional.

The term *safety CPLD* describes the device's assigned role, not a certification. A watchdog inside that CPLD does not independently cover failure of the same device. Board design addresses its power, reset, configuration, clock, and stale-permission failures, with bounded safe response and relay de-energization independent of continued logic progression. Figure 5.3 shows two possible implementations.

5.7 Safe First, Diagnose Second

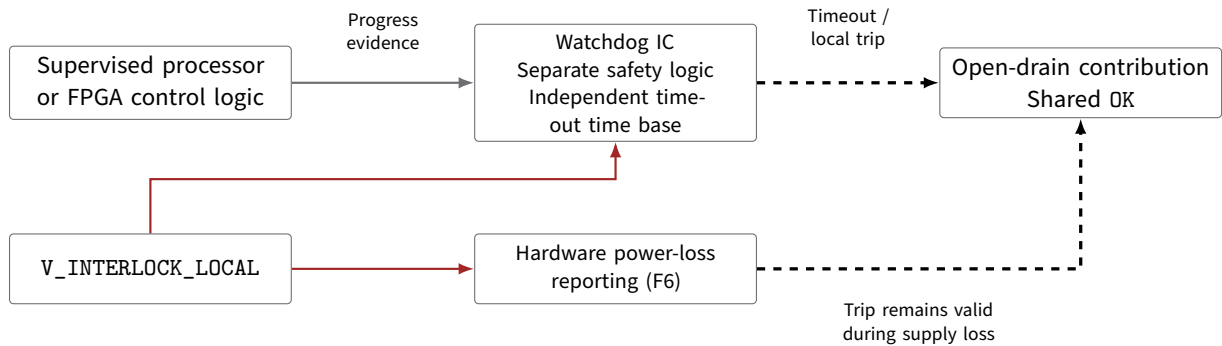
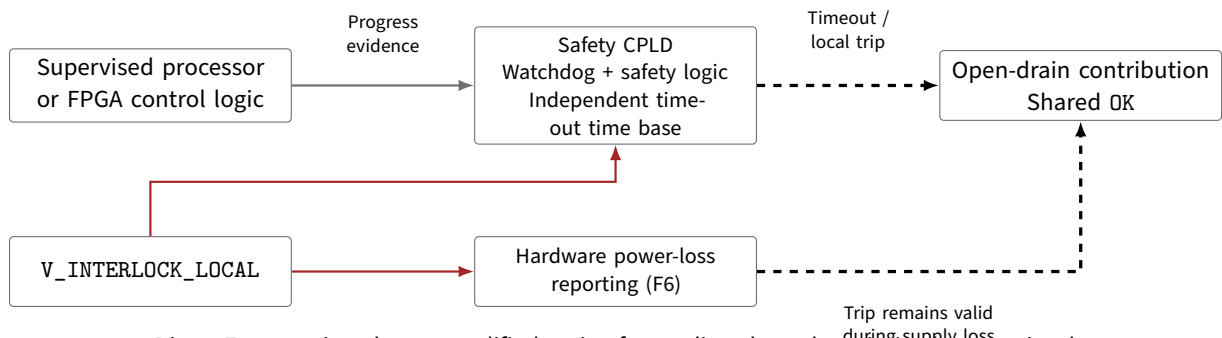
The clock monitor and watchdog report separate failures. Acquisition-clock loss enters the clock-monitor trip path. Missing processor/control-logic refreshes instead cause watchdog timeout. Either path can make the system safe before host diagnosis.

Each board retains local evidence that helps the host distinguish events after the trip. Clock-loss evidence is separate from watchdog activation. A responding board can report whether it detected a local fault, a host-supervision timeout, or only the shared trip from another participant. Main retains evidence of a broken continuity loop. Common-rail measurements help identify a power fault.

Watchdog activation directs investigation to processor/control-logic execution and its refresh path, separately from acquisition-clock loss. Communication silence alone establishes only that the board is unreachable, not that its power or logic has failed. Some severe failures leave a board unable to communicate. The host then combines the missing response with the hardware trip and the reports from the remaining boards. Diagnosis may be less precise, but the safe response does not depend on diagnostic completeness.

5.8 Trips Remain Latched Until Explicit Recovery

Restoring a clock, rail, temperature, or communication link does not release a retained trip. The host must request recovery, followed by live-condition qualification and a new arm request. Chapter 6 explains the sequence.

A: Separate watchdog device**B: Combined safety functions**

Direct F6 reporting shown; qualified active forwarding through main is also permitted.

Both examples require fail-safe power/reset behavior and relay de-energization without continued logic progression.

A CPLD watchdog does not independently cover failure of that same CPLD; board design must address it.

Figure 5.3: Alternative implementations of the same safety functions. Both preserve independent watchdog timing and safe failure behavior. Direct F6 reporting is shown; qualified active loop forwarding is another permitted F6 path.

5.9 Finding a Damaged Fault-Output Path

An open-drain fault driver can fail open. In that condition, the board may appear healthy during normal operation but fail to pull OK low when later needed. The continuity loop does not detect this electronic failure.

The system therefore supports a maintenance action that commands one selected board to assert its OK path while the host verifies that the shared bus changes state. The independent hardware watchdog path is also tested safely. Tests remain safely disarmed and demonstrate that the selected contribution causes an observable HIGH-to-LOW transition. A bus already held LOW by another source makes the assertion test inconclusive. Active protection is never bypassed to obtain a baseline.

The system ICD and maintenance plan define commands, schedule, acceptance timing, and service procedure.

Figure 5.4 illustrates the observable test behavior without fixing command names or timing values. Release follows the defined safe test/recovery behavior; full system recovery is not necessarily needed solely to release a test contribution. Real fault retention and recovery still apply. Testing one contribution does not validate every independent watchdog or supervisor path.

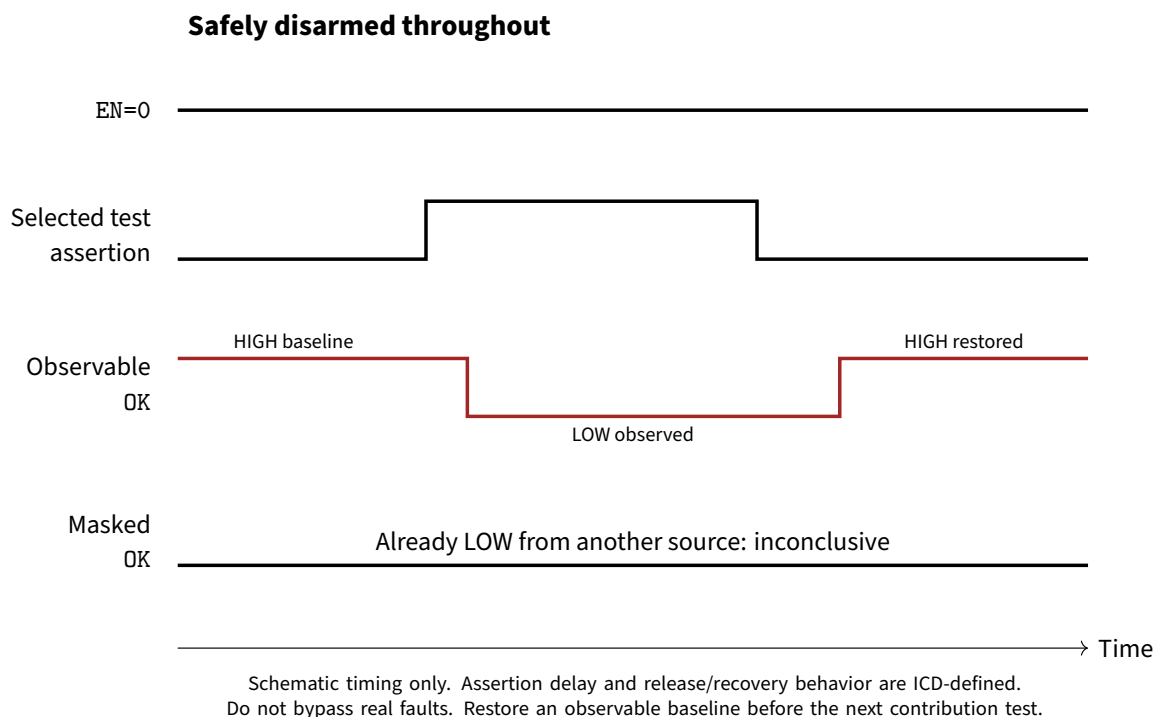


Figure 5.4: Conclusive and masked assertion tests while safely disarmed. Time is schematic; acceptance intervals and safe test/recovery sequencing belong to the ICD and maintenance plan.

5.10 Safety Scope

The shared interlock provides a common response to detected faults. It does not prove that every possible detector hazard has been identified. Each function-board design still needs a hazard analysis for its detector-facing voltages, clocks, stored energy, output sequencing, and failure modes.

The architecture does not select watchdog parts, relay types, voltage thresholds, hold-up components, FPGA latches, or diagnostic registers. It also does not claim a formal safety integrity level. Those choices and claims require board-level analysis, verification, and any project-specific certification process.

The value of the architecture is a clear safety boundary: detector-facing permission is normally absent, independent hardware can remove it, all boards share the safe response, and software cannot override an active trip.

Chapter sources: ADR-001 defines the fault classes and independent detection paths. ADR-003 defines the safety layers, relay permissive, trip retention, and recovery rules. ADR-004 defines independent clock monitoring, and ADR-005 defines shared-rail fault reporting.

6 Operating Lifecycle and State Coordination

The controller contains several independent boards, but an instrument must still start, arm, acquire data, stop, and recover as one system. The architecture provides a common externally visible lifecycle for this purpose.

The lifecycle does not require every board to use identical FPGA/CPLD code or identical internal states. It defines what the host and the other boards can observe. Each board remains responsible for its own readiness, local work, and fault evidence. The main board coordinates only the shared actions.

6.1 Distributed Coordination

The host, main board, and function boards have separate responsibilities:

- The **host** knows the instrument configuration, sends operational settings, requests system actions, and gathers status.
- The **main board** drives the shared EN, CLEAR, CLOCK, and SYNC signals and observes shared health.
- Each **function board** checks its own readiness, controls its detector function, and enters the common safe response when required.

Before arming, the host verifies that board configuration and readiness match the intended operation. For example, a host may configure sequences and bias or clock voltages while disarmed, disconnect, and be replaced by another host. The new host may use the retained configuration after verifying that it matches the intended operation; a change of host does not itself require a reload or an interlock trip. The ICD defines the verification method.

There is no separate backplane readiness bus. The host reads readiness over the network. During arming, each function board also checks its own conditions before it allows local detector-facing permission.

This arrangement keeps coordination common without making the main board responsible for the internal operation of every function board.

6.2 The Four State Families

Every active board presents behavior consistent with four state families: START, IDLE, RUN, and ERROR. Figure 6.1 shows the main transitions.

The state names describe system behavior rather than one mandatory state encoding. A simple board may implement fewer internal steps. A complex video or clock board may need extra local substates. Both are acceptable if their external behavior matches the common lifecycle.

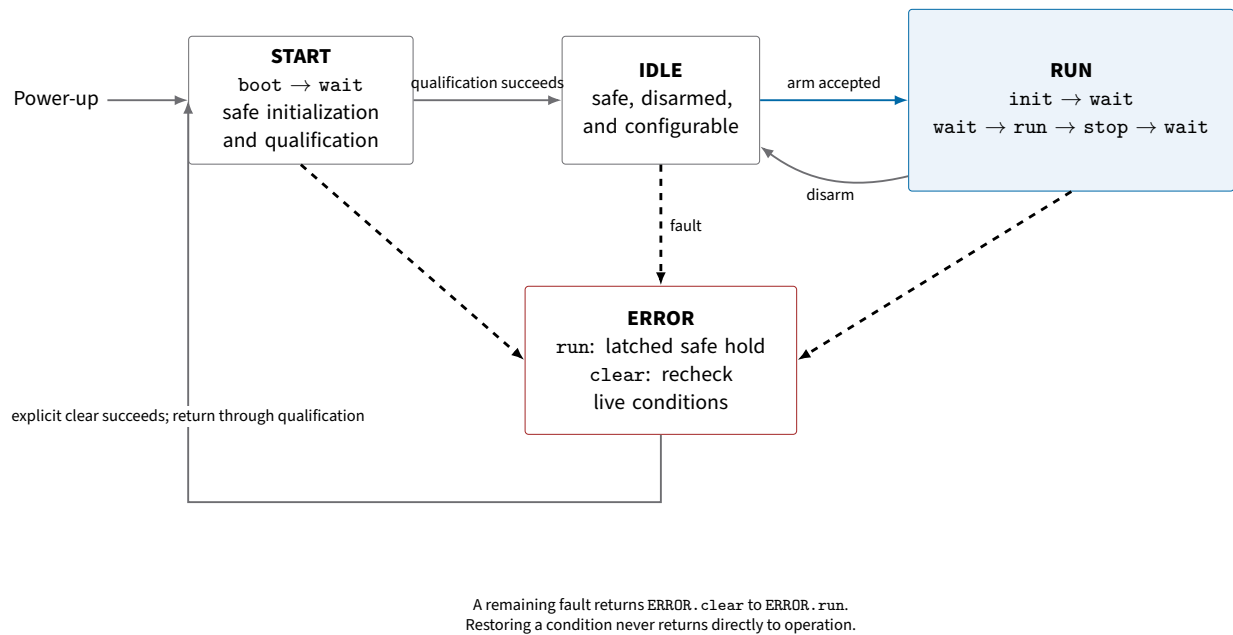


Figure 6.1: Common state families and externally visible transitions. Internal implementation may differ between boards.

State family	Meaning
START	The board initializes in a safe condition and then qualifies the power, timing, communication, monitoring, and shared health that apply to its role.
IDLE	The board is safe, disarmed, and available for allowed configuration. Detector protection relays remain open.
RUN	The controller is armed. The board prepares for acquisition, waits for timing events, performs its local work, and completes any controlled stop work.
ERROR	The board is held safe with fault evidence available. Recovery requires an explicit request and a new qualification pass.

Table 6.1: Externally visible meaning of the common state families.

6.3 Safe Startup and Qualification

At power-up, boards enter `START.boot`. Detector-facing outputs remain disabled. Each board establishes safe defaults, reads its identity, starts communication, and enables local monitoring.

The board then enters `START.wait`. In this stage it checks the live conditions needed for safe operation. Depending on the board role, these checks may include stable shared health, valid local power, required timing activity, watchdog operation, and communication readiness.

Startup qualification is bounded. A missing clock or a rail that never becomes valid cannot leave the system waiting forever and still appear usable. The applicable ICD defines the deadline and the required stable-health interval.

Boards may hold the shared OK bus low while their independent protection paths initialize. The system cannot reach the armable condition until required participants have released their contributions and health remains stable. A known local safety fault sends the affected board to `ERROR` instead of `IDLE`.

Successful qualification leads to `IDLE`. This is the normal safe state for configuration and inspection.

6.4 Configuration in IDLE

Operational settings are loaded while the controller is safely disarmed. Examples include detector biases, clock parameters, acquisition settings, enable masks, and sequencer data. These values are current-session information sent over Ethernet. They are separate from persistent board identity and bootstrap-network data.

Sequencer payloads are also volatile. A sequencer board remains Not Ready until the host completes the required loading transaction. If the payload is later changed, the board returns to Not Ready until the new transaction completes.

This leads to an important distinction:

Condition	Meaning while disarmed	Result
Not Ready	A required operational step is incomplete, but no hazardous failure is present	Arming is blocked; the board remains safe and can normally be configured
Fault	A safety-related hardware or electrical condition has been detected	The board asserts its trip as required and enters the latched safe error state

Table 6.2: Not Ready prevents operation; Fault requires the interlock response.

A board that has not received a sequencer is Not Ready, not necessarily faulty. A missing required rail or a failed watchdog is a fault. Keeping these meanings separate avoids turning ordinary configuration work into a hardware trip.

Operational writes are rejected outside the safe states defined by the communication and board ICDs. Reset or power loss restores safe defaults and clears volatile sequencer readiness. A normal arm and disarm cycle does not by itself erase valid current-session settings.

6.5 Distributed Arming

The host begins arming by sending a request to the main board. Main accepts the request only from IDLE and only after its own requirements are satisfied. It then raises the shared EN signal.

Raising EN is not enough to force a function board to operate. Each function board independently checks its applicable conditions. These can include valid local settings, loaded sequencer data, stable power and timing, no retained fault, and a valid hardware interlock.

If the checks pass, the board enters `RUN.init` and prepares its local function. It then asserts its own local arm request and reaches `RUN.wait`. Only the combination of local permission, EN, and OK can energize the detector protection relay described in Chapter 5.

If a board detects an unsafe arm attempt, it uses its normal local trip path. The controller returns to the shared safe response. This is stronger than allowing the system to run with one required board silently unarmed.

The main board waits through an arm-to-first-trigger guard before sending the first acquisition event. This gives every installed board enough time to recognize EN and complete its verified `RUN.init` work. The ICD defines the guard from the slowest supported board behavior.

6.6 Acquisition Within RUN

The RUN family separates arming from an individual acquisition:

- `RUN.init` performs local arm-entry preparation;
- `RUN.wait` remains armed and waits for an acquisition event;
- `RUN.run` performs the timing-sensitive acquisition work; and
- `RUN.stop` completes the board-specific controlled stop before returning to `RUN.wait`.

A rising acquisition SYNC event starts the acquisition action. A falling event stops it. The exact detector clocks, sample sequence, buffering, and data framing remain local to the applicable function boards.

After `RUN.stop`, the controller may remain armed in `RUN.wait` for another acquisition. It does not need to return to IDLE between every exposure or readout.

Acquisition traffic may also provide the qualifying host interaction described in Chapter 8.

Figure 6.2 shows the main interactions in a normal cycle. It represents one logical sequence, not exact protocol messages.

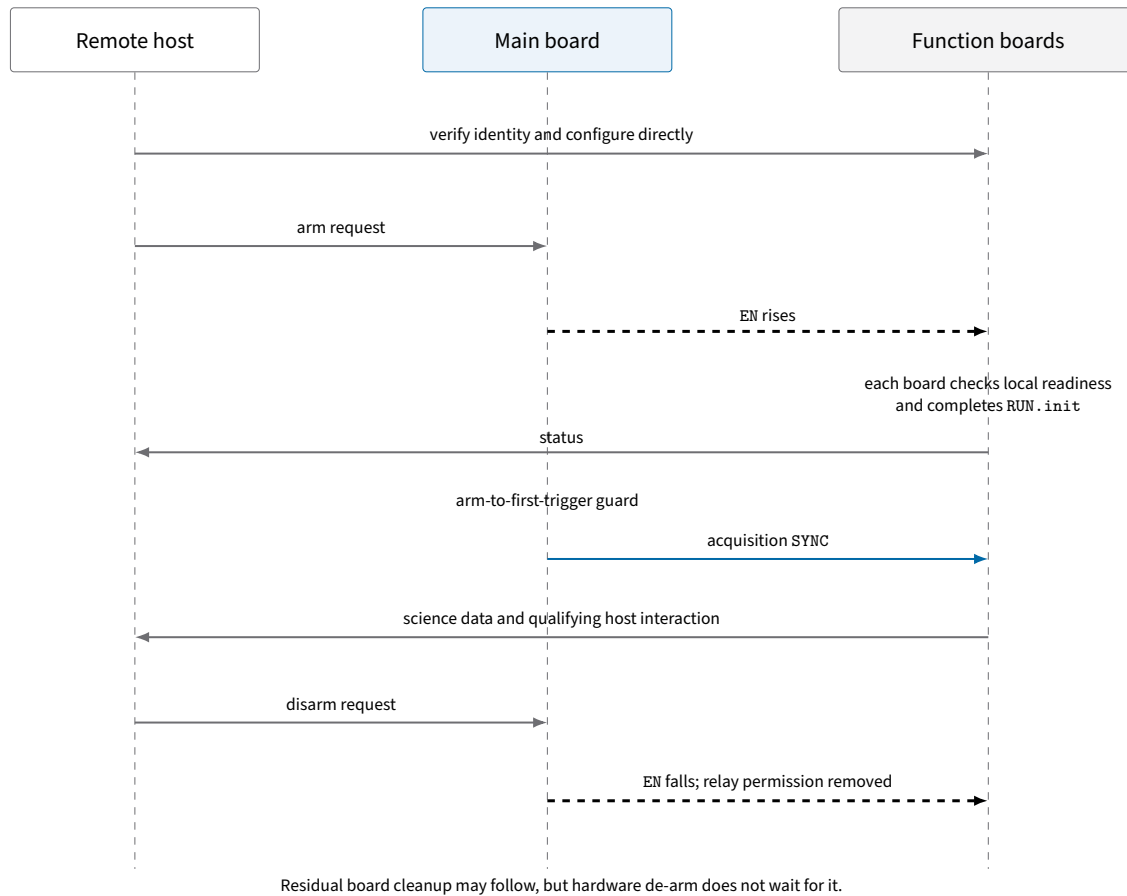


Figure 6.2: Normal configuration, arm, acquisition, and disarm sequence. Host communication with function boards is direct.

6.7 Hardware-First Disarm

To disarm, the host asks the main board to drop system enable. This differs from stopping an acquisition while remaining armed. Main drops `EN`. The falling level removes function-board relay permission through hardware. This action does not wait for a processor response or for the board FSM to complete cleanup.

Boards then perform any remaining local cleanup and return to `IDLE`. The difference is important: the hardware becomes safe first, and orderly software or firmware cleanup follows.

6.8 Fault Entry

A detected local fault causes the affected board to assert its contribution to `OK` and retain evidence. The low shared bus removes detector-facing permission across the controller. Boards enter `ERROR.run`, where they remain safe and make diagnostics available when communication still works.

A board may enter `ERROR` because another participant tripped. It does not pull `OK` low merely because it observed the shared bus low. If it later detects its own local fault, however, it records and asserts that fault as well.

Fault entry has priority over commanded operation in `START`, `IDLE`, or any `RUN` substate. Chapter 5 describes the hardware paths that make this response independent of normal state progress.

6.9 Explicit Recovery

Recovery begins only after the controller is safe and the cause has been investigated. The host sends a clear request to the main board, and main drives the shared `CLEAR` signal.

Each locally faulted board enters `ERROR.clear` and checks its live conditions. `CLEAR` is a request, not a command to declare health. A board whose fault remains active keeps its trip asserted and returns to `ERROR.run`.

When local clearing succeeds, the system returns through `START.wait`. Required timing and shared health are qualified again before `IDLE`. There is no direct path from `ERROR` to `IDLE`, and there is never an automatic return to `RUN`.

Valid current-session operational settings do not have to be erased by ordinary fault recovery. A board may require reconfiguration when its local failure or reset invalidated those settings. Its readiness status tells the host what must be loaded again before the next arm.

6.10 Armed Host Supervision

Continued armed operation with an unavailable or unresponsive host is not assumed safe. A board need not determine whether the host froze or its communication path failed before initiating a shared trip.

Host supervision is active while $EN=1$. Every active board, including main and function boards, requires a qualifying bidirectional host interaction within a bounded interval. If the interval expires, that board records a supervision event, asserts its local trip, and places the system in the safe error state.

While disarmed ($EN=0$), loss of host contact alone does not assert OK. Hardware protection remains active. Data overrun alone also does not cause a trip. Chapter 8 explains qualifying interactions and their coexistence with science traffic.

6.11 Timing Categories Belong in ICDs

The lifecycle requires bounded timing, but this handbook does not assign numerical values. The system and board ICDs define the values from verified hardware and operational needs.

The main categories include startup deadline, stable-health qualification, clock qualification, clear pulse and recovery time, arm-to-first-trigger guard, host-supervision timeout, watchdog timing, and any required converter settling time.

Keeping the categories in the architecture ensures that important behavior is bounded. Keeping the values in ICDs allows them to change when hardware and instrument requirements become known without rewriting the system architecture.

Chapter sources: ADR-003 defines the common state families, arming, fault entry, disarm, and recovery. ADR-002 defines configuration and sequencer readiness. ADR-004 defines acquisition timing ownership, and ADR-006 defines data-path interaction with armed host supervision.

7 Timing Domains and Synchronization

The controller uses several clocks and synchronization signals for different jobs. They should not be treated as one timing system.

Detector sequencing needs a common, deterministic clock. Acquisition start and stop events must be associated with the same clock cycle on every participating board. Management and safety must continue if that distributed timing is lost. Power converters use another synchronization mechanism to make switching noise predictable.

Figure 7.1 separates these functions.

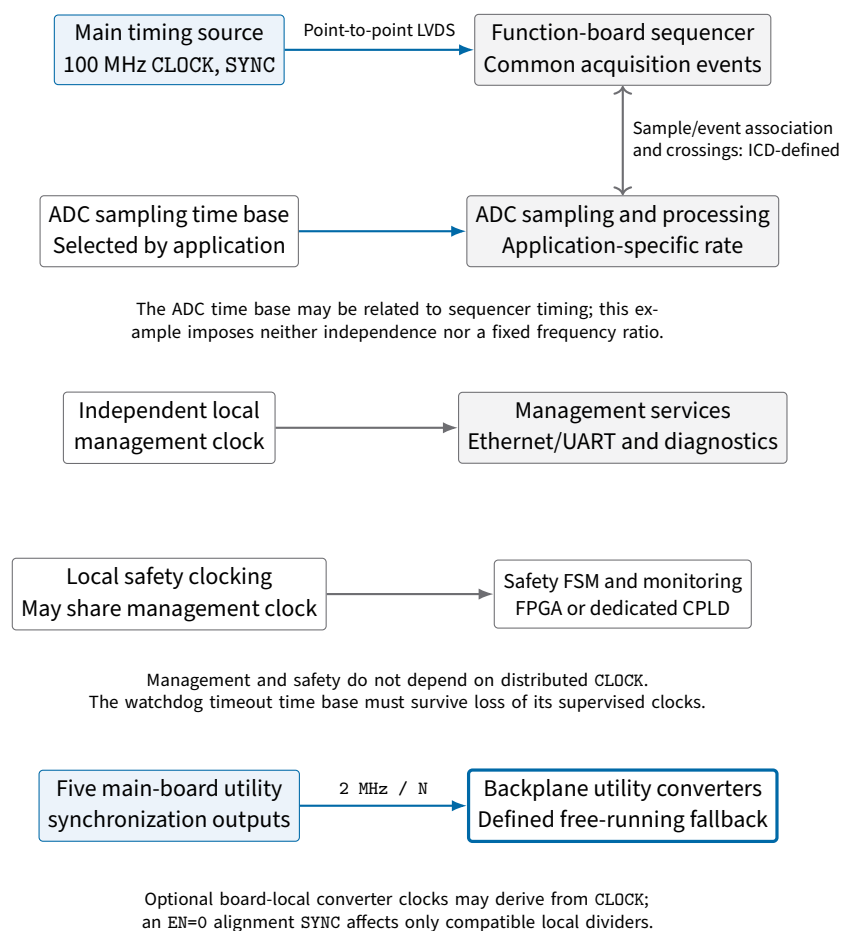


Figure 7.1: Timing-domain relationships. The sequencer path is coherent across boards, local management remains independent, and utility-converter synchronization is a separate function.

7.1 The Distributed Sequencer Clock

The main board distributes a continuous 100 MHz `CLOCK` for detector sequencing. Function boards use the full-rate clock directly for timing-critical work. They may divide it by an integer for slower synchronous functions, but they do not recreate the 100 MHz clock by multiplying a slower reference.

Using the same full-rate source gives participating boards a common timing reference. A clock board can generate detector phases while a video board associates samples and processing windows with sequencer events. The exact local actions remain board-specific.

The architecture does not select the clock source, fanout device, receiver, or FPGA I/O method. Those choices must meet the instrument timing budget.

ADC sampling rate is application-specific; the 100 MHz sequencer clock does not impose a universal ADC rate. The video-board ICD defines the sampling time base and its relationship to acquisition events and processing windows, including any required clock-domain crossings. An ADC maximum supported rate is distinct from the selected operating rate.

7.2 Point-to-Point Slot Distribution

Each supported slot receives its own LVDS `CLOCK` pair and its own LVDS `SYNC` pair from the applicable fanout. The timing interface is point-to-point rather than a multidrop transmission line.

This avoids slot stubs and population-dependent loading. An empty slot does not change the electrical load seen by a populated slot. It also lets the timing design verify each source-to-receiver path with a clear endpoint.

Point-to-point does not mean that every path is automatically identical. Fanout delay, trace length, connectors, receivers, and environmental changes still contribute to the total timing budget. The ICD defines the electrical interface and the allowed path differences.

When a bridge extends timing to another backplane, the extension must still present point-to-point timing to each supported slot. The instrument design verifies all added propagation, skew, and clock-quality effects. The architecture does not require one universal multi-backplane topology.

7.3 Acquisition SYNC and Cycle Agreement

The `SYNC` signal carries acquisition start and stop events. A rising event starts acquisition behavior, and a falling event stops it. These events are timing-critical only where they enter the sequencer domain.

Main changes `SYNC` on a falling edge of the distributed 100 MHz `CLOCK`. Participating function boards capture the received level on the following rising edge. This nominal half-cycle separation prevents a transition from arriving close to the active edge that captures it. Without this rule, small path differences could make one board associate an event with the previous cycle while another chooses the next cycle.

The 180-degree relationship is therefore about selecting the same intended cycle. It is not a general requirement to phase-align every clock in the controller. The timing ICD still verifies the actual setup and hold margin after all source, fanout, routing, connector, receiver, skew, and environmental effects.

Timing-critical ADC controls, such as conversion-start signals, remain in deterministic sequencer-domain paths. Their exact FPGA implementation and constraints belong to the video-board design.

7.4 Independent Management Clocking

Each active board uses local clocking so management and safety functions can continue when the distributed acquisition clock disappears. The board can then detect timing loss, assert its local trip, retain diagnostics, communicate with the host, and participate in recovery. Management serves host supervision, diagnostics, Ethernet, and UART independently of acquisition timing.

Management may also need to know that acquisition has started or stopped. It receives the SYNC level or acquisition-event information through a reliable clock-domain crossing. Unlike the sequencer, management does not need to observe the event on the exact acquisition-clock cycle, so its clock does not require phase alignment with the distributed clock.

For example, the sequencer acts on SYNC at the specified clock edge; management subsequently records that acquisition has started. The reporting delay does not change when the sequencer acted. This management copy serves status and diagnostics, not timing-critical acquisition control.

A board may align a local clock for implementation convenience, but safety, communication, fault reporting, and recovery must remain available without that alignment or the distributed clock. Operational timeouts use the appropriate local time base; oscillator accuracy and timer values belong to the applicable ICDs.

Safety functions may share the management clock or use a separate local safety clock, including one in a dedicated CPLD. The watchdog timeout time base must remain operational when the clocks it supervises stop. A common oscillator for all these functions is not required.

7.5 Clock-Loss Detection and Watchdog Evidence

Clock loss and frozen processing are separate failures. A clock monitor detects loss of acquisition timing, while an independent watchdog detects when the processor or control logic stops refreshing it. Either can assert OK LOW and make the system safe, without waiting for the other or for the host.

On a function board, the clock monitor observes the distributed 100 MHz CLOCK; on main, it observes the external clock source. It uses independent local management or safety clocking, so it can still detect the loss when the acquisition clock stops. A disconnected clock cable therefore produces a clock-loss trip.

The processor or control logic periodically refreshes the watchdog as part of its normal execution. These refreshes are also called *petting*. If execution freezes and refreshes stop, the watchdog times out and asserts its hardware trip. Refreshes use local clocking. An automatic pulse generator that keeps refreshing the watchdog after the processor freezes would defeat this protection.

A board may optionally align refresh transitions with the acquisition clock to make their switching activity more predictable for noise control. If the external clock disappears, refresh timing uses the local clock. If it returns, alignment may resume. These changes must not cause a false timeout or allow refreshes to continue without processor/control-logic execution. The watchdog's own timeout time base remains independent throughout.

For example, a disconnected acquisition-clock cable causes a clock-loss trip. A frozen processor that stops refreshing its watchdog causes a watchdog trip. Each failure has its own detector and diagnostic record.

Watchdog protection applies while armed and disarmed, with startup qualification preventing arming before protection is ready. Host-supervision timeout remains a separate rule that trips only while armed.

The board records clock loss and watchdog timeout separately. After a trip, the host uses the clock-loss record to investigate acquisition timing and the watchdog record to investigate processing and its refresh path. If both are recorded, both conditions were detected; the records do not establish their order. If the board is unreachable, its local records are unavailable. The shared safe response does not wait for diagnosis.

7.6 Timing Loss and Recovery

Loss of required acquisition timing trips the controller and removes detector-facing permission as described in Chapter 5. The board management domains continue to operate.

Restoring the 100 MHz clock does not resume acquisition. The system remains in the safe error state until the host requests recovery. Boards clear retained evidence only at the allowed recovery point, requalify live timing, return to IDLE, and wait for a new arm request.

This rule prevents a loose cable, unstable source, or intermittent receiver from restarting detector activity as soon as a waveform happens to return.

7.7 Utility-Converter Synchronization Is Separate

The backplane utility converters do not use the 100 MHz detector sequencer interface. Main provides five separate point-to-point LVDS outputs for utility-converter synchronization.

During acquisition, each enabled switching channel uses a frequency of $2 \text{ MHz} / N$, where the positive integer N and channel mapping are defined by the backplane ICD. Converter phase behavior is also an ICD decision. Forced continuous, fixed-frequency operation keeps the switching pattern predictable.

If synchronization is absent, the utility converter remains enabled and free-runs in a defined fixed-frequency continuous mode. This fallback is different from loss of the required detector sequencer clock. It does not create an uncontrolled rail or require processor intervention.

The converter synchronization settings remain fixed during acquisition. Chapter 4 explains the noise-control reason for this choice.

7.8 Optional Board-Local Converter Synchronization

A function board may synchronize one or more compatible local converters when coherent switching improves detector performance. Their continuous switching references may be derived from the distributed CLOCK and, when appropriate, use the same $2 \text{ MHz} / N$ convention.

Whenever the system enters *START*, main sends one converter-alignment SYNC pulse while *EN*=0. A compatible function board recognizes any such pulse whether its own state is *START*, *IDLE*, or *ERROR*. It may then reset or align its local converter divider. The pulse does not clear a fault or change the board's safety state.

A board that requires this alignment completes any required settling before declaring itself ready. With *EN*=1, SYNC has acquisition meaning only and does not reset local converter timing. Boards that do not use this feature ignore the alignment pulse.

This use is optional and local to the board. Divider, phase, settling, filtering, and free-running fallback belong to the board ICD and design.

Optional converter timing does not change the acquisition SYNC capture rule and does not create another shared system state.

7.9 Timing Budgets Belong to the Application

The architecture defines which relationships must be controlled, but it does not set universal jitter, skew, or delay numbers. The acceptable values depend on the detector, ADC, FPGA, board count, routing, and selected clock components.

The timing ICD verifies the complete path from the source through fanout, backplane routing, connectors, receivers, FPGA logic and I/O, data converters, and detector interface circuits. A multi-backplane instrument includes the bridge and extension path in the same analysis.

This approach keeps the important concept fixed: all participating boards must select the intended acquisition cycle. The numerical budget remains with the interface and implementation that can be measured and verified.

Chapter sources: ADR-004 defines sequencer timing, acquisition SYNC, independent management clocks, clock-loss evidence, and multi-backplane timing. ADR-001 defines watchdog and fault evidence. ADR-003 defines acquisition and recovery behavior, and ADR-005 defines utility-converter synchronization.

8 Identity, Communication, and Data Architecture

Every active board is an independent network device. The host must know which board performs each instrument role, configure it safely, supervise it while armed, and receive its data. This must still work when a board is replaced or its network settings are wrong.

The architecture separates persistent service information from normal operation. UART provides local commissioning and recovery. Ethernet provides operational control, status, supervision, and science data. The host owns the instrument inventory.

8.1 Direct Host-to-Board Communication

Each active board has its own Ethernet endpoint. The host communicates directly with the main board, every function board, and every bridge board. Function boards do not send ordinary traffic through the main board, and they do not command one another.

The main board is one endpoint among the others. It receives system coordination requests, such as arm, disarm, clear, and acquisition timing requests. A video board receives its own configuration and sends its own science data. A clock or bias board receives the settings for its local function.

This direct model avoids two bottlenecks. Main does not need to process all board control traffic, and it does not need to carry the combined data rate of every video board. Communication capacity can grow with the installed boards and the instrument network.

Figure 8.1 shows the communication ownership and the separate UART service path.

The architecture does not require one physical switch arrangement. Boards may connect directly or through suitable network switches. The instrument ICD defines the topology and verifies that each link has enough capacity for its control, telemetry, supervision, configuration, and acquisition traffic.

8.2 Information and Configuration

Not all board information should be stored or changed in the same way. Table 8.1 groups the main information classes by ownership, channel, and lifetime.

This separation prevents an ordinary operational command from silently changing the identity or future network behavior of a board. It also prevents old detector settings from returning automatically after a reset.

8.2.1 Operational Configuration

The host sends normal operating values over Ethernet while the board is safely disarmed. The board validates the values before applying them. The communication and board ICDs define which writes are allowed in each state and how invalid requests are reported.

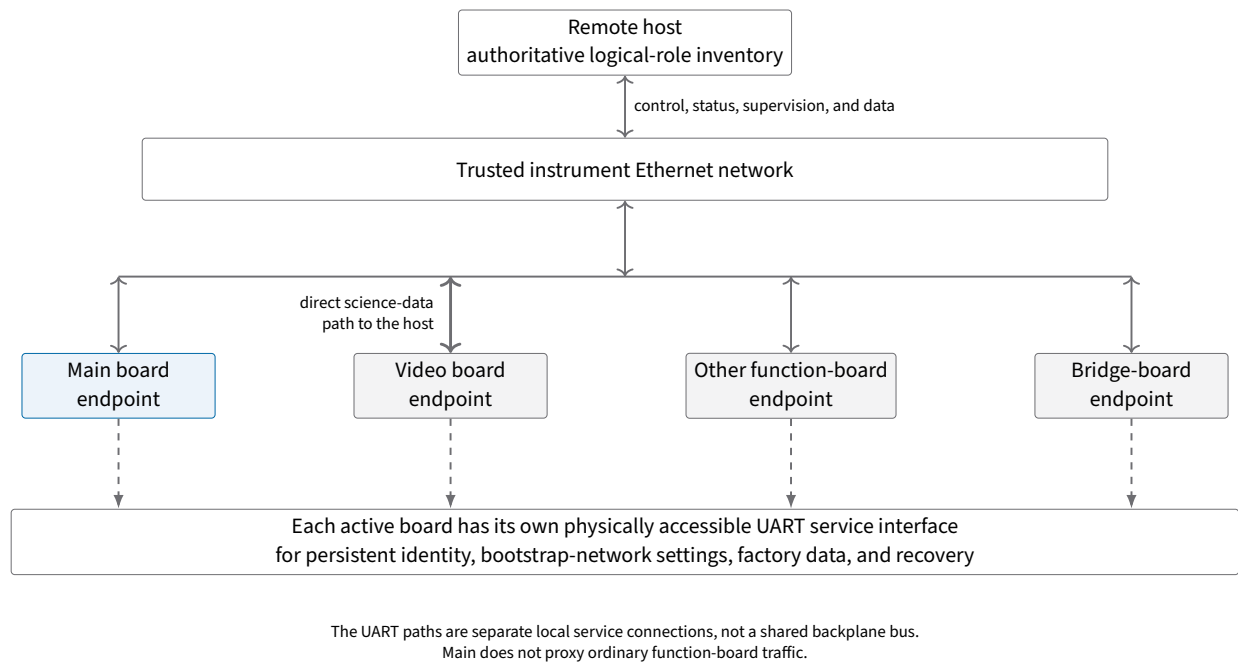


Figure 8.1: Every active board is a direct Ethernet endpoint. Each also has its own local UART service interface for persistent commissioning and recovery.

Information	Owner or source	Normal channel	Lifetime
Board identity and bootstrap network	Board persistent storage; commissioned through service tools	UART service interface	Persists across reset and power loss
Factory calibration or trim	Board persistent storage and manufacturing records	UART service interface	Persists across reset and power loss
Firmware version	Active firmware image	Ethernet normally; UART during service	Describes the running image
Operational settings	Host sends values to the board	Ethernet	Current boot session
Sequencer payload	Host transfers to the sequencer board	Reliable ICD-defined host transfer	Current boot session; readiness is cleared by reset or power loss
Diagnostics and telemetry	Board monitoring and retained evidence	Ethernet normally; UART for recovery	Observation; selected evidence may remain until recovery
Science data	Video function board	Direct board-to-host data path	Acquisition product managed by the host system
Instrument inventory	Remote host	Host configuration and commissioning records	Persists independently of any one controller board

Table 8.1: Information classes use channels and lifetimes that match their purpose.

Operational values are volatile. Reset or power loss restores safe defaults. Fault recovery and ordinary arm or disarm cycles do not automatically erase valid settings from the current boot session.

This behavior supports efficient recovery without treating potentially stale values as permanent board configuration. A board that lost or invalidated part of its operational state reports Not Ready until the host restores what is needed.

8.2.2 Sequencer Transfer and Readiness

Sequencer payloads are loaded only while safely disarmed. The transfer uses a reliable protocol defined by the applicable ICD. A sequencer board remains Not Ready until the host explicitly completes the transaction.

If accepted sequencer content is later modified, readiness is cleared again. Reset or power loss also clears sequencer readiness. A normal fault clear or arm/disarm cycle does not have to erase a valid payload unless the board's local behavior invalidated it.

The readiness indication confirms completion of the host loading transaction. The architecture does not require or prohibit an additional hash, content signature, or attestation check. Those mechanisms may be added by the board or instrument specification.

8.3 Service, Identity, and Board Replacement

Ethernet cannot be the only way to repair Ethernet configuration. A stale static address, incorrect bootstrap setting, or damaged operational image could make a board unreachable through its normal network path.

Every active board therefore provides a physically accessible UART service interface. It supports commissioning, allowed persistent-data access, fallback diagnostics, and recovery when normal Ethernet startup fails.

The UART connection is local to each board. It is not a shared backplane inventory bus. This keeps recovery available without adding another multi-board addressing system.

The communication ICD defines the electrical standard, connector, framing, request and response behavior, validation, and recovery entry. The architecture only requires that the service path remain accessible and independent of normal network reachability.

Interrupted persistent-data writes must not create a partially valid record. The board hardware and firmware designs choose the storage technology, validation method, redundancy, and atomic update method.

8.3.1 Persistent Board Identity

Each active board provides the persistent information needed to identify and commission it. Where applicable, this includes:

- a unique MAC address;
- a unique serial or board identifier;
- board type and hardware revision;
- bootstrap IP information when static addressing is used; and
- required manufacturing calibration or trim data.

Software version is not stored as board identity. It is reported by the active firmware image. This allows the host to distinguish the physical board from the software currently running on it.

Normal detector biases, clock settings, timing values, enable masks, acquisition settings, and sequencer content are not persistent identity. They are operational values loaded for the current session.

Ethernet does not modify persistent board storage. Changes to identity, bootstrap settings, or factory information use the controlled UART service path.

8.3.2 Host-Owned Logical Inventory

Physical slot position is not the operational identity of a board. A controller may have different slot populations, and a bridge may extend the system to another backplane. Tying a logical role only to a slot would make replacement and topology changes difficult to manage.

The host owns the authoritative mapping:

logical instrument role $\longleftrightarrow$ board capability and revision $\longleftrightarrow$ network endpoint $\longleftrightarrow$ MAC and serial identity

At startup, the host checks that each expected endpoint reports the expected identity and capability. A network response alone is not enough. The wrong board type at a reachable address must not be accepted as the required instrument function.

Slot information may still help a technician locate hardware, and an identification light or discovery tool may assist commissioning. Neither replaces the host inventory.

The main board does not need this complete function-board inventory to distribute shared electrical signals. It provides timing and coordination to the supported interfaces independently of the logical role assigned by the host.

8.3.3 Replacing a Board

A replacement board is commissioned in two deliberate steps.

First, a technician uses the board's UART service interface to verify its physical identity and set permitted bootstrap-network information. Second, the host inventory is updated to map the intended logical role to the replacement board's identity and endpoint.

The host then verifies the new mapping during startup and loads the required operational settings. Copying an old IP address is not enough, because the serial identity, hardware revision, or capability may differ.

The architecture does not define a complete service procedure. It defines the information and independent recovery path needed for a safe procedure to be written.

8.4 Supervision and Science Traffic

While armed, every active board requires a valid host interaction within a bounded interval. The interaction must prove communication in both directions. A board that only sends unsolicited data cannot know whether the host still receives or responds to it.

A host status request and board response can provide both acquisition-time health visibility and the qualifying supervision exchange. This demonstrates communication activity, not correct operation of every part of the host application. Measurements help the host diagnose conditions; protection requiring a timely local response must not depend on the host reading telemetry and sending a stop command.

The architecture does not mandate a separate heartbeat message. Telemetry, a heartbeat, an acknowledgement, data credit, or another bidirectional operation may qualify. The communication ICD selects the operation, cadence, retries, response, and timeout.

Chapter 6 describes the armed timeout and fault response.

The implementation is verified at maximum supported traffic. A full data queue or a busy acquisition task must not delay the qualifying interaction without bound.

8.4.1 Science Data Goes Directly to the Host

Each video function board sends acquisition data directly to the remote host over its own network endpoint. There is no acquisition-data bus on the backplane, and the main board never carries science data.

The video-board and system ICDs define link rate, transport, framing, integrity checks, buffering, flow control, and reconnect behavior. The architecture does not choose TCP, another transport, a frame format, or a universal Ethernet speed. A board may use 100 Mb/s, 1 Gb/s, or another supported capacity according to its application. The application defines data products and rates, local processing or reduction, and buffering, with capacity verified at maximum supported load while preserving host supervision.

Using one board endpoint for both control and data avoids a second connector and physical link before measurements show that one is needed. The link budget must include maximum science traffic together with control, status, and supervision traffic.

If a specific instrument cannot meet its measured data requirement with that link, its ICD and design may add a dedicated data link. The direct board-to-host ownership and host-supervision rules remain unchanged.

8.4.2 Data Overrun Is Not Automatically a Safety Fault

If the network or host cannot accept data fast enough, a video board may overrun a buffer or lose part of an acquisition. This damages data quality, but it does not by itself mean that detector interface electronics

are unsafe.

The board reports enough information for the host to identify the affected frames or acquisition. The transport follows its ICD-defined recovery behavior. The board does not pull OK low solely because data was lost.

Host supervision remains independent. If the board cannot complete the required bidirectional host interaction within the allowed armed interval, the supervision timeout still trips the controller. Local buffering and task scheduling must be sized and verified so maximum acquisition traffic does not hide this timeout path.

8.5 Trusted Instrument-Network Boundary

The standard architecture assumes a physically or logically isolated instrument network with trusted peers. This is a deployment condition, not a statement that Ethernet traffic is safe on any open network.

An instrument that cannot provide this boundary adds authentication or other controls in its deployment and communication specifications. Message-integrity checks may also be used where they improve reliability, independently of authentication.

A complete cybersecurity design depends on the observatory network, operating environment, update process, and threat model. Those details are outside this architecture handbook.

Chapter sources: ADR-002 defines persistent identity, UART recovery, host-owned inventory, operational configuration, direct board endpoints, and sequencer readiness. ADR-003 defines legal operating states and supervision consequences. ADR-006 defines direct acquisition-data ownership and overrun behavior.

Part III

Applying the Architecture

9 Operating and Fault Scenarios

The previous chapters explained the architecture one topic at a time. This chapter follows complete operating stories across power, safety, state, timing, communication, and data.

Each scenario illustrates the architecture under a particular operating condition. All recoverable faults use the common recovery process at the end of this chapter.

Each scenario asks the same questions: What was the initial condition? What changed? How was it detected? What happened in hardware? What state followed? What evidence remained? What is required before operation can resume?

9.1 Normal Startup, Acquisition, and Disarm

The controller begins unpowered and unarmed. When external power is applied, the central eFuse qualifies the input and supplies protected +12V. The backplane utility converters start, and their rail-health outputs keep the shared system from becoming armable until the common rails are valid.

Each active board starts from safe defaults. Its detector protection relays remain open. The board establishes identity, communication, monitoring, local interlock power, and its state logic. It then checks the timing, power, shared health, and local conditions that apply to its role.

The host connects directly to the active boards and verifies each identity against the instrument inventory. It sends operational settings and loads required sequencer payloads while the system is in IDLE. Boards report Ready only when their applicable configuration and live conditions are valid.

Before arming, the host verifies that configuration and readiness match the intended operation. The host then asks main to arm. Main raises EN; each function board performs its own acceptance checks and completes `RUN.init`. After the arm-to-first-trigger guard, main may send acquisition SYNC. Function boards perform their local timing work, and video boards send science data directly to the host.

For a normal stop, main sends the falling SYNC event and boards complete their controlled acquisition stop. The controller may remain armed for another acquisition. When the host requests disarm, main drops EN. Hardware relay permission disappears before any remaining firmware cleanup. Boards return to safe IDLE.

While disarmed, host disconnection alone does not assert OK; hardware protection remains active. If another host connects, it may use retained settings and sequences after verifying that configuration and readiness match the intended operation. Changing hosts does not itself require a reload or an interlock trip.

Figure 6.2 shows the normal sequence.

9.2 Arm Attempted While One Board Is Not Ready

Assume the controller is safely in IDLE, but one required sequencer board has not completed its loading transaction. The board is Not Ready. It has not detected a hardware fault, so it does not pull OK low merely because configuration is incomplete.

The host should see the readiness status and avoid the arm request. If an arm is attempted anyway, main may know only its own readiness and the shared health state. It raises EN only if its local checks pass.

The Not Ready function board then performs its independent arm acceptance. Because a required condition is missing, it does not request local relay permission. An unsafe arm attempt uses the board's normal trip path, pulling OK low and placing the fleet in the safe error state.

Diagnostics show that the initiating board rejected the arm because it was not ready. The host clears the error through the normal recovery path, loads the missing sequencer, verifies readiness, and arms again. There is no separate shared readiness wire and no way for the host to force the local board to accept the arm.

9.3 Three Different Power-Loss Cases

Power faults can look similar to an operator because detector-facing outputs turn off in every case. Their detection and remaining diagnostic capability differ. Figure 9.1 compares the three main cases.

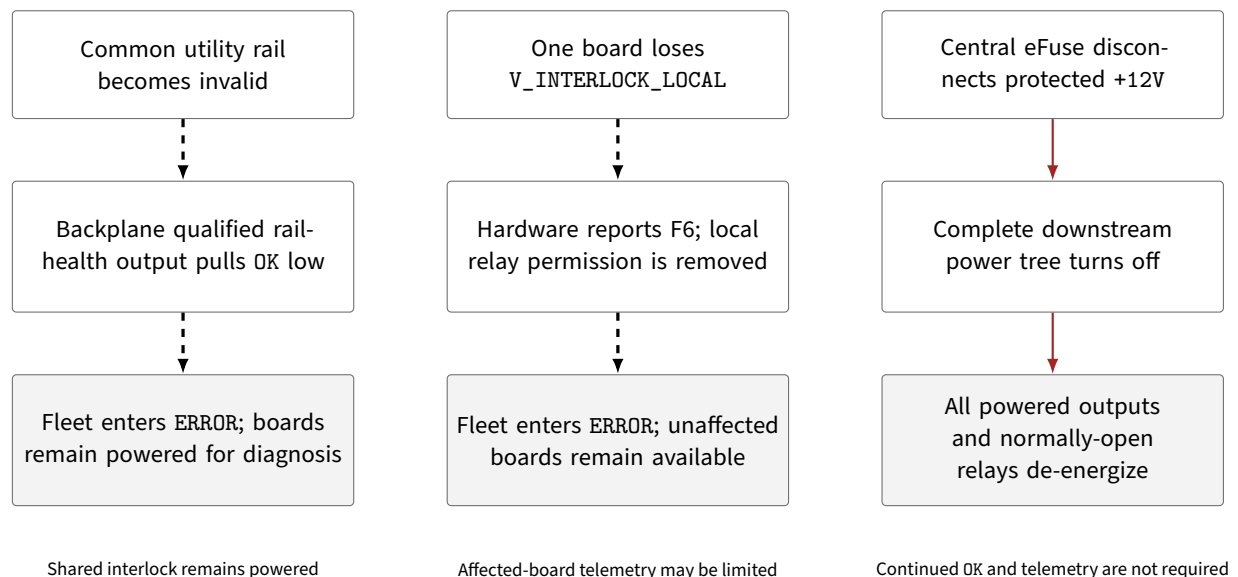


Figure 9.1: Comparison of a common utility-rail fault, isolated loss of one board's local interlock power, and complete central eFuse cutoff.

9.3.1 A common utility rail becomes invalid

The controller is powered, possibly armed, when one backplane utility rail leaves its valid range. Its qualified rail-health output pulls OK low directly. The hardware permissive opens detector protection relays, and active boards enter ERROR.

Protected +12V remains present, so boards normally continue to communicate. Main measures the common rails, and the host combines those measurements with board diagnostics. The common recovery process applies even if the voltage returns by itself.

9.3.2 One board loses local interlock power

Assume the shared power system remains valid, but one board loses V_INTERLOCK_LOCAL because of a connector branch or local regulator failure, or because an optional board input eFuse disconnects the branch supplying interlock power. The board causes a shared trip either through a direct hardware contribution to OK or through qualified active forwarding that makes main's loop return LOW. Its normally-open detector protection relay also releases as local interlock power collapses. Downstream boards using incoming-loop validity as a hardware permissive can remove local permission before main asserts the shared trip.

The rest of the controller remains powered and enters ERROR. Diagnostics from the affected board may be limited if more of its local power is also lost. Other boards and main still provide evidence. If the board became physically disconnected, the continuity loop may report that condition as an additional cause.

9.3.3 The central eFuse removes complete system power

An unsafe +12V_IN condition, excessive total current, or protected +12V outside its verified range causes the central eFuse to disconnect protected +12V. Utility converters and all active boards turn off. Normally-open relays de-energize directly.

The shared OK bus and network telemetry do not need to survive this complete cutoff. Safety comes from removing the energy that powered the outputs. Diagnosis then uses the eFuse indication, external supply evidence, and inspection after power is restored. A normal interlock trip does not command this cutoff; it is a separate input-protection action.

9.4 Digital Power Collapse Versus Frozen Logic

Two failures can make a board stop responding while leaving different hardware evidence.

If the processor or FPGA rail collapses, or the FPGA is held in reset, its normal logic cannot be trusted to assert a fault. The fail-safe circuit uses V_INTERLOCK_LOCAL and treats loss of the active health signal as unsafe. It pulls OK low without waiting for the watchdog.

If the logic remains powered but freezes, a stale health output might remain in its previous state. The independent hardware watchdog detects the missing pet transitions and pulls OK low. Its output path does not depend on the frozen logic.

In both cases, the fleet enters the same safe response. The difference is diagnostic. A board that still communicates may report watchdog activation or another retained event. A completely unresponsive board is reported as unreachable; investigation considers power, logic, and communication failure without assigning a cause from silence alone. Safe action is not delayed while the host makes that distinction.

9.5 Distributed Sequencer-Clock Loss

Assume the controller is acquiring data when a 100 MHz CLOCK path becomes invalid. The cause could be the source, main-board fanout, bridge path, connector, receiver, or local timing logic.

An affected board observes clock activity using its independent local management or safety clock. It records clock-loss evidence and uses its local trip path to pull OK low. Every function board loses hardware permission and enters the safe error state.

The retained clock-loss record directs investigation to the clock source and distribution. Watchdog timeout is a separate fault associated with processor/control-logic execution and its refresh path.

Figure 9.2 shows the order. Exact detection and watchdog delays are intentionally not shown.

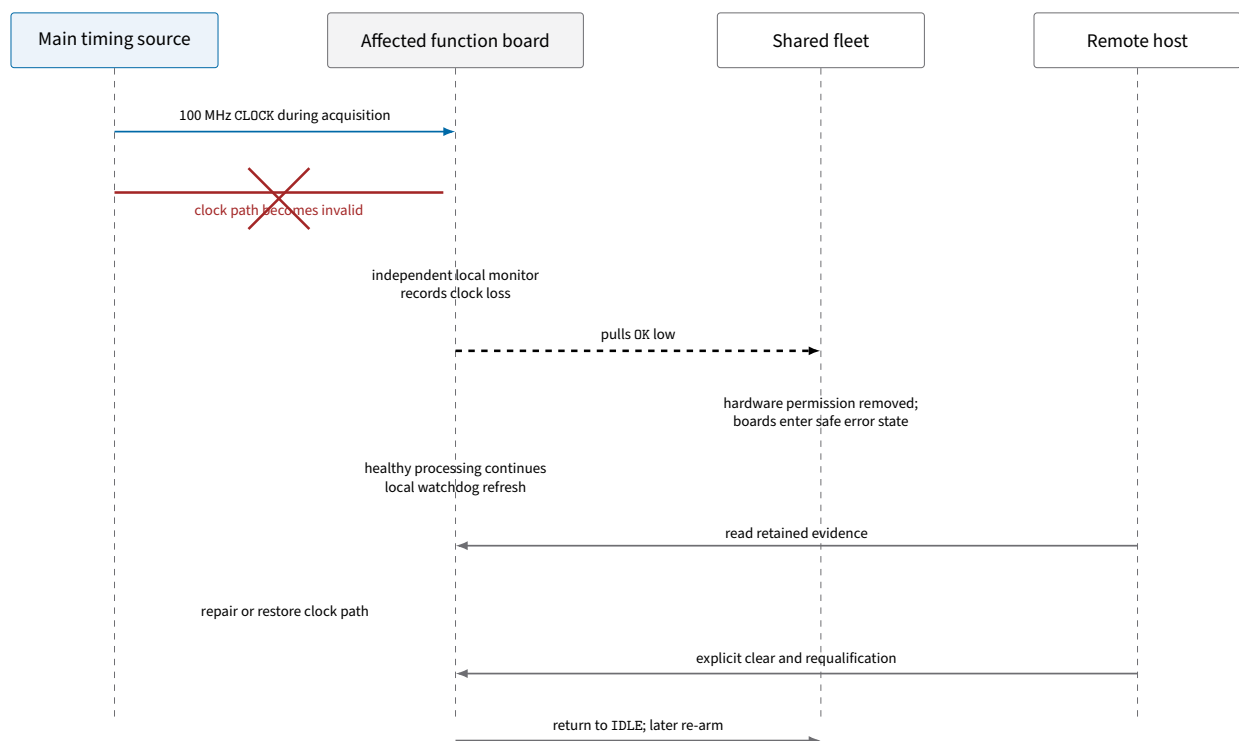


Figure 9.2: Clock-loss response, diagnosis, and recovery. Restoring timing does not restart acquisition automatically.

The host reads evidence from responding boards to determine whether all paths lost timing or only one branch did. After repair, the common recovery process includes timing requalification.

9.6 Host Communication Loss During Acquisition

Assume the controller is armed and acquiring data when host communication is interrupted. The interruption may affect one board, one switch path, or the complete instrument network; the host application itself may also have frozen. Continued armed operation with an unavailable or unresponsive host is not assumed safe, so boards need not distinguish these causes before taking the protective response.

Every active board maintains its own armed supervision interval. A qualifying operation must demonstrate communication in both directions. A host telemetry request and board response may qualify, providing health visibility during acquisition without a separate heartbeat message. A video board continuing to transmit frames has not, by that fact alone, proved that the host is still reachable.

When the first affected board fails to complete its required interaction within the allowed interval, it records the supervisory event and pulls OK low through its normal local trip path. Hardware relay permission is removed across the fleet, and boards enter ERROR.

After communication returns, the host gathers evidence and follows the common recovery process at the end of this chapter.

9.7 Data Overrun Without Supervision Timeout

Now assume the host and board still complete their qualifying bidirectional interactions, but the network or host cannot absorb all science data. A video buffer overruns or a frame becomes incomplete.

The board reports a data-quality event and identifies the affected acquisition as defined by its ICD. It does not pull OK low solely because science data was lost. Detector safety and data completeness are different concerns.

The acquisition transport follows its defined overrun behavior. It may stop, discard, mark, or recover data according to the instrument design. If valid host interaction continues, the armed supervision timer remains satisfied.

If congestion also prevents the qualifying interaction from completing before its deadline, the independent host-supervision rule applies and the board trips the system. Maximum-load testing must show that this safety interaction cannot be starved indefinitely by science traffic.

9.8 Physical Discontinuity

Assume a board, passive terminator, bridge cable, or connector opens the continuity path. The missing or unpowered item cannot be relied upon to report its own removal.

The continuity path is interrupted, so `LOOP_IN` at main becomes invalid. Deterministic main-board logic retains evidence and asserts its contribution to `OK`. Downstream function boards that use incoming-loop validity as a hardware permissive can remove local detector-facing permission before main asserts `OK`. The shared `OK` trip then places the entire powered fleet in the safe error state, including boards upstream of the interruption. This safe-state response does not itself force loop outputs `LOW`.

If a bridge board loses interlock power, qualified active forwarding makes the loop return `LOW`; a bridge with passive copper instead reports through a separate hardware contribution to `OK`. A `LOW` loop return alone does not distinguish physical disconnection from power loss in an active forwarder. If the cable is physically severed, the loop detects the open path. Both pieces of evidence may appear when one event causes several failures.

An active forwarder may also make its output `LOW` to report a detected condition that prevents local safety hardware from performing its role. Downstream boards propagate `LOW`, and main initiates the shared trip. This is an additional reporting path, not a claim that every internal device failure is detected or that the affected board can be located from the return alone. Forwarding resumes when the local condition clears and power/configuration are valid, without waiting for shared-trip recovery; detector outputs still require the normal recovery and arming process.

Removing or inserting boards is therefore not a normal live operation. Service is performed while powered down or already latched safe. The damaged-`OK`-driver maintenance check described in Chapter 5 covers a different latent electronic failure.

9.9 Common Scenario Summary

Table 9.1 collects the main distinction between the scenarios. It shows the first architectural response, not every later diagnostic possibility.

9.10 Generic Fault Correction and Recovery

Every recoverable fault scenario returns through the same high-level process.

First, hardware has already removed detector-facing permission and the controller is held in `ERROR.run`. The host reads the evidence that remains available. The operator or control system corrects the external or local condition. Correction may mean restoring a clock, repairing a cable, replacing a board, fixing power, or restoring network service.

Condition correction does not release retained trips by itself. The host requests clear. Each affected board enters `ERROR.clear`, checks the live condition, and releases its own trip only if allowed. A remaining fault returns that board to `ERROR.run`.

When clearing succeeds, the controller returns through `START.wait`. Shared health, timing, monitoring, communication, and applicable board readiness are qualified again. The system then returns to safe `IDLE`.

Scenario	First detection or decision	Architectural result
Board Not Ready at arm	Function-board local arm acceptance	Unsafe arm is rejected; local trip places fleet in ERROR
Common utility-rail fault	Backplane qualified rail health	OK low; fleet safe while protected bulk power normally remains
One local interlock supply lost	Hardware power-loss reporting and relay disable	OK low directly or through main; local relay releases; fleet safe
Complete eFuse cutoff	Central input protection	Complete power tree and normally-open relays de-energize
Processor/FPGA power or reset collapse	Interlock-powered fail-safe path	OK low without relying on the failed logic
Powered logic freeze	Independent hardware watchdog	OK low after missing valid pet activity
Distributed clock loss	Independent-clock activity monitor	OK low; clock and watchdog evidence remain separate
Armed host timeout	Board-local supervision timer	Board uses its normal local trip path
Data overrun only	Acquisition buffering and transport logic	Data-quality report; no trip solely for the overrun
Physical discontinuity	Continuity loop and main-board conversion	Main pulls OK low

Table 9.1: Representative scenarios and their first architectural response.

Any operational settings or sequencer content invalidated by the event are loaded again. The host verifies the instrument inventory and that board configuration and readiness match the intended operation. A new arm request is required before detector-facing outputs can be enabled.

This common recovery path is deliberate. It avoids special automatic-restart rules for different faults and ensures that restoration is followed by a complete check of the conditions needed for safe operation.

Chapter sources: The scenarios combine ADR-001 through ADR-006. ADR-001 controls fault detection, ADR-003 controls state response and recovery, ADR-004 controls timing-loss behavior, ADR-005 controls the power cases, ADR-002 controls readiness and host interaction, and ADR-006 controls data overrun and acquisition traffic.

Part IV

Reference Appendices

A Glossary

This glossary reproduces the repository's controlled vocabulary so readers can interpret recurring terms consistently. The README remains authoritative; this table is generated during the publication build.

Preferred term	Meaning
Main board	Board that drives shared coordination signals such as EN, CLEAR, CLOCK, and SYNC
Function board	Active plug-in board performing detector-specific work, such as video, bias, or clock generation
Bridge board	Active board extending supported signals to another backplane
Passive terminator	Unpowered slot insert that maintains continuity-loop routing
Backplane board	Physical PCB carrying slots, shared signals, protected +12V, and common utility converters
Active board	Powered main, function, or bridge board participating in communication or safety logic
Utility rail	Mandatory common low-current rail generated on the backplane board
Qualified rail-health output	Open-drain backplane contribution to OK for a shared rail, provided by a verified native converter PGOOD or a dedicated voltage supervisor
V_INTERLOCK_LOCAL	Low-power board-local supply for independent interlock hardware; not a backplane-distributed rail
PGOOD	A native power-good output from a power converter; the term is not used for watchdog or fail-safe outputs
+12V_IN	External system power before the central backplane eFuse
+12V	Protected bulk rail after the central eFuse; distributed to the backplane loads and active boards
Fault	Hardware or electrical condition requiring safe-state action
Supervisory interlock event	Non-hardware condition, such as armed host-supervision timeout, requiring the same safe response
Not Ready	Condition that blocks arming while disarmed but is not itself a fault
Trip	Action that asserts the interlock, normally by pulling OK LOW
Recovery	Explicit ERROR.clear → START.wait → IDLE process
Host inventory	Mapping of logical role to board type/revision, IP, MAC, and serial ID
Topology	Physical signal-distribution structure, such as an LVDS star or multi-backplane clock tree

Use **board** when the distinction among main, function, and bridge boards is unimportant. Use **module** only for a complete removable mechanical assembly when that distinction matters.

B ADR Traceability Index

This index connects the explanatory chapters to the ADRs that control their technical content. It is generated from structured comments in each chapter. It indicates topic coverage, not formal requirement-by-requirement compliance.

B.1 Handbook Chapter to ADR

Handbook chapter	Controlling ADRs
Chapter 1 — Origins and Guiding Ideas	ADR-001, ADR-002, ADR-003, ADR-004, ADR-005, ADR-006
Chapter 2 — The System at a Glance	ADR-001, ADR-002, ADR-003, ADR-004, ADR-005, ADR-006
Chapter 3 — Modular System Structure	ADR-001, ADR-002, ADR-003, ADR-004, ADR-005, ADR-006
Chapter 4 — Power, Supervision, and Noise Architecture	ADR-001, ADR-003, ADR-005
Chapter 5 — Hardware Safety and Interlock Architecture	ADR-001, ADR-003, ADR-004, ADR-005
Chapter 6 — Operating Lifecycle and State Coordination	ADR-002, ADR-003, ADR-004, ADR-006
Chapter 7 — Timing Domains and Synchronization	ADR-001, ADR-003, ADR-004, ADR-005
Chapter 8 — Identity, Communication, and Data Architecture	ADR-002, ADR-003, ADR-006
Chapter 9 — Operating and Fault Scenarios	ADR-001, ADR-002, ADR-003, ADR-004, ADR-005, ADR-006

B.2 ADR to Handbook Chapter

ADR	Explanatory chapters
ADR-001	Chapter 1 , Chapter 2 , Chapter 3 , Chapter 4 , Chapter 5 , Chapter 7 , Chapter 9
ADR-002	Chapter 1 , Chapter 2 , Chapter 3 , Chapter 6 , Chapter 8 , Chapter 9
ADR-003	Chapter 1 , Chapter 2 , Chapter 3 , Chapter 4 , Chapter 5 , Chapter 6 , Chapter 7 , Chapter 8 , Chapter 9
ADR-004	Chapter 1 , Chapter 2 , Chapter 3 , Chapter 5 , Chapter 6 , Chapter 7 , Chapter 9
ADR-005	Chapter 1 , Chapter 2 , Chapter 3 , Chapter 4 , Chapter 5 , Chapter 7 , Chapter 9
ADR-006	Chapter 1 , Chapter 2 , Chapter 3 , Chapter 6 , Chapter 8 , Chapter 9

The source comments and generated tables are publication metadata. The ADR files remain the unique source for architectural requirements. Chapter-end source notes also identify relevant project-context documents and repository guidance where needed.

C Shared-Signal Summary

The table below is extracted from ADR-003 and provides a compact view of the shared coordination signals. It summarizes architectural behavior only. Electrical levels, pin assignments, biasing, termination, pulse widths, timing margins, and inter-backplane implementation belong to the applicable ICDs.

Signal	Driver and topology	Architectural behavior
OK	Open-drain contributions from boards and qualified backplane rail-health sources; pull-up on main	HIGH means no participant is asserting a fault. Any participant may pull LOW. A LOW trip has priority over commanded operation.
LOOP_OUT / LOOP_IN	Series continuity path originating and terminating at main; passive copper or unconditional active forwarding	A broken return path causes main to assert its OK fault contribution.
EN	Main; shared level	HIGH only while the system is armed in RUN. LOW means safe/disarmed. Function boards independently qualify a rising edge; falling removes hardware permission without waiting for an FSM transition.
CLEAR	Main; shared level/pulse	Requests locally faulted boards to clear retained evidence as appropriate and re-evaluate live conditions. It does not force a board to declare itself healthy.
CLOCK	Main clock source; point-to-point LVDS	Continuous 100 MHz sequencer clock. Function boards use it directly for timing-critical work and independently detect its loss.
SYNC	Main; point-to-point LVDS	With EN=1, rising and falling acquisition events are captured in the 100 MHz domain. With EN=0, a pulse is a converter-alignment event that compatible function boards recognize in any local state. A separate CDC-safe observation is used for management. Detailed behavior is defined by ADR-004.

Table C.1: Shared-signal behavior extracted from ADR-003 R3 during the publication build.

Five point-to-point utility-converter synchronization pairs, numbered 0 through 4, are also part of the standard main-board/backplane interface. They synchronize backplane utility-converter channels during acquisition. Their signal names, mapping, divisors, phase behavior, and electrical implementation are defined by the backplane ICD under ADR-005.

A function board may additionally derive compatible local converter clocks from its point-to-point CLOCK. A SYNC pulse received while EN=0 may align those local dividers in any local board state. With EN=1, SYNC has acquisition meaning only. This optional board-local use is defined by ADR-004 and does not consume or redefine the five backplane utility-converter synchronization pairs.

Board-local watchdog, fail-safe, relay-control, and voltage-supervisor signals are not shared backplane signals. Their internal names and implementation belong to board design specifications.

D Power-Rail Summary

The power architecture separates the external input, protected bulk distribution, common utility rails, and board-local supplies. This appendix is a navigation aid; ADR-005 and ADR-001 remain authoritative.

D.1 System Input and Protected Bulk Rail

Rail	Owner/location	Purpose
+12V_IN	External input to backplane	Unprotected system input before the central eFuse
+12V	Backplane output after central eFuse	Protected bulk supply for utility converters and active boards; source for high-current and specialized board-local conversion

The central backplane eFuse is the system input circuit breaker. It disconnects if protected +12V cannot remain within the verified input range of downstream converters. Its detailed thresholds, current limits, delays, ratings, supervision method, and retry or latch behavior are implementation decisions. A normal interlock or common utility-rail trip removes detector-facing permission but does not command the eFuse to remove system power.

D.2 Mandatory Common Utility Rails

Rail	Intended use
+6V_ANA	Common positive low-voltage analog utility rail
-6V_ANA	Common negative low-voltage analog utility rail
+16V_ANA	Common positive analog utility rail
-16V_ANA	Common negative analog utility rail

The utility-rail rows are extracted directly from ADR-005 during the build. A standard backplane generates the complete set even when an installed board does not consume every rail. Exact voltage tolerances, capacity, ripple, transient, connector, and per-slot limits belong to the backplane ICD.

D.3 Board-Local Supplies

Supply class	Architectural role
V_INTERLOCK_LOCAL	Low-power supply derived on every active board from protected +12V. It powers only the independent watchdog, fail-safe OK path, power-loss protection hardware, and relay-reset support where applicable. It is not distributed by the backplane.
Digital point-of-load rails	All ordinary digital supplies, including identification, monitoring, management, processor, FPGA, and memory rails, generated locally from protected +12V. The standard backplane does not distribute a low-voltage digital utility rail.
Specialized detector rails	Non-common voltages generated locally when required by a detector-facing board and controlled by that board's ICD and design specification.
Analog conditioning and local regulation	Filters, LDOs, or other circuits used to meet incoming-noise and returned-disturbance limits at a shared utility-rail boundary.

Shared rail health is owned by the backplane and contributes directly to OK. The main board measures common voltages for diagnosis. Board-local supervision is added only where a safety-relevant function is not already covered by the fail-safe or watchdog paths.

E Official Project References

The following project documents provide provenance, historical context, and the controlled-document-format example. Resolved ADRs remain authoritative for the current architecture.

NOIRLab Detector Group (Aug. 15, 2025). *Obsolescent Detector Controller Replacement Strategy Study: Project Report*. Official project source document in the repository. NOIRLab.

- (Feb. 16, 2026). *NOIRLab Detector Controller: Architecture Overview*. Official project-context document in the repository; resolved ADRs control where descriptions differ. NOIRLab.

NOIRLab-ES Detector Group (Sept. 9, 2025). *Obsolescent Detector Controller Phase II: Project Mandate*. Version 1.0. Official project source document in the repository; internal-use distribution. NOIRLab.

These bibliography entries identify repository-held project documents rather than external architecture authorities. Their dates and descriptions preserve provenance; later resolved ADRs control current technical behavior where wording differs.

F Version and Generation Information

Handbook version	0.2
Publication status	Working version
Generation date	September 9, 2026
Architecture authority	Resolved Architecture Decision Records in the source repository

This appendix is generated from publication metadata and the build date. It does not define architecture.

F.1 Source Hierarchy

Repository area	Role
decisions/	Resolved ADRs containing the current cross-system architectural decisions
interfaces/	Planned ICDs containing exact boundary definitions and implementation-specific values
design/	Planned board hardware and firmware design specifications
integration/	Explanatory and integration material
publication/	Derived LaTeX handbook sources, generators, diagrams, and build tooling

The version and generation date identify the PDF. For a controlled release, the approved PDF and its matching repository snapshot are archived under the same formal document version.